Introduction to Quantum Physics: A Lecture Course

PHYS 214, Spring 2026, ZJUI Institute (Prof. Said Mikki)

Contents

1 Course Scope and Preliminary General Considerations

Course Roadmap

- **Lecture 1**: Course introduction and overview. Big picture of quantum mechanics: universal scope and multidisciplinary applications. First look at the two approaches to QM: quantization versus the abstract algorithmic view.
- **Lecture 2**: Mathematical foundations for quantum mechanics. Review of waves, complex numbers, probability, and basic linear algebra. Introduction to vectors in Hilbert space and Dirac notation.
- **Lecture 3**: Historical approach to quantum mechanics. Blackbody radiation, photoelectric effect, and Bohr's model. Introduction of de Broglie waves and Heisenberg uncertainty principle.
- **Lecture 4**: Conceptual foundations through the double-slit experiment. Explanation using wave mechanics and discussion of wave-particle duality. Introduction of the probability interpretation and superposition.
- **Lecture 5**: Foundations of wave mechanics I. Postulates of QM in Schrödinger picture: wavefunctions, operators, and the Schrödinger equation. Measurement, expectation values, and the meaning of the wavefunction.
- **Lecture 6**: Foundations of wave mechanics II. Solving the Schrödinger equation in 1D: free particle, infinite potential well, finite well, and particle in a box. Introduction to wave packets.
- **Lecture 7**: Dirac-Hilbert formalism I. Moving from wavefunctions to abstract state vectors in Hilbert space. Introduction of Dirac notation, spin as a qubit system, and quantum measurement in the abstract picture.
- **Lecture 8**: Dirac-Hilbert formalism II. Completeness, basis expansions, and operators in the abstract framework. Unification of wave mechanics and the abstract formalism.
- **Lectures 9-10**: Harmonic oscillator. Both analytic solution (Hermite polynomials) and algebraic method (ladder operators). Importance in physics and quantum field theory.
- **Lectures 11-12**: Atoms, molecules, and solids. Hydrogen atom, angular momentum, and spin. Pauli exclusion principle, multi-electron atoms, and introduction to band theory in solids.
- **Lectures 13-14**: Multi-particle systems, entanglement, and applications. Tensor products, identical particles (bosons and fermions), quantum entanglement, and introduction to quantum information.

1.1 Course Overview and Objectives

Course Overview and Objectives

This course introduces the fundamental principles of quantum mechanics, the theoretical framework that describes physical systems at atomic and subatomic scales. By the end of this course, you will be able to:

- Understand the historical and experimental motivations for quantum theory
- Master the mathematical formalism of quantum mechanics (Hilbert spaces, operators, eigenvalues)
- Solve the Schrödinger equation for key potentials (infinite well, harmonic oscillator, hydrogen atom)
- Apply quantum mechanics to atomic, molecular, and solid-state systems
- Develop intuition for quantum phenomena (superposition, entanglement, measurement)

Prerequisites: Classical Mechanics, Electromagnetism, Linear Algebra, Differential Equations.

1.2 Math background Needed

Key Mathematical Tools You Will Need

- **Differential equations**: Solving the Schrödinger equation
- **Linear algebra**: Vector spaces, eigenvalues, inner products
- **Complex analysis**: Complex number algebra, Euler identity.
- **Probability**: Elementary concepts.
- **Analysis**: Basic function theory, calculus in several dimensions, elementary Fourier transforms.

1.3 Engineering Audience Targeted in this Course

Four Undergraduate Disciplines Served

This course is designed to serve students from four major engineering and science disciplines:

Computer Science (CS)
Quantum computing, algorithms

Electrical Engineering (EE)
Quantum devices, nanofabrication

Computer Engineering (CE)
Hardware-software integration

Mechanical Engineering (ME)
Nanomechanics, quantum materials

1.4 Two Major Application Categories

Dual Focus of Quantum Applications

1. Quantum Information Processing
(CE, EE, CS focus)

2. Physical Systems
(ME, CE, EE focus)

a. Materials Science
Quantum properties of solids, fluids, semiconductor materials, metamaterials

b. Quantum Devices
Electron/phonon devices, nanorobots, nanomaterials and systems

Examples of Applications You Can Work on In the Future Opened Up by This Course

Computer Science (CS) Focus

- **Quantum algorithms**: Shor's, Grover's, quantum machine learning
- **Quantum complexity theory**: BQP, QMA complexity classes
- **Quantum programming**: Qiskit, Cirq, quantum circuit design
- **Quantum error correction**: Surface codes, fault tolerance
- **Cryptography**: Post-quantum cryptography, quantum key distribution

Electrical & Computer Engineering (EE/CE) Focus

- **Quantum hardware**: Qubit implementation (superconducting, spin, topological)
- **Control systems**: Microwave pulse engineering, cryogenic electronics
- **Nanofabrication**: Josephson junctions, quantum dot devices
- **Photonic integration**: Quantum photonics, integrated optics
- **System architecture**: Quantum-classical interface, readout electronics

Mechanical Engineering (ME) Focus

- **Quantum materials**: 2D materials, topological insulators, superconductors
- **Nanomechanics**: Optomechanical systems, NEMS/MEMS quantum sensors
- **Thermodynamics**: Quantum heat engines, nanoscale thermal transport
- **Fluid dynamics**: Quantum fluids, superfluids, Bose-Einstein condensates
- **Solid mechanics**: Quantum effects in nanomaterials, strain engineering

1.5 Cross-Disciplinary Applications

Application Area	**CS**	**CE**	**EE**	**ME**
Quantum Computing	✓	✓	✓	
Quantum Sensing		✓	✓	✓
Quantum Materials			✓	✓
Quantum Communication	✓	✓	✓	
Quantum Simulation	✓		✓	✓
Quantum Metrology		✓	✓	✓

Unifying Theme

Despite different applications, all disciplines share the **same mathematical foundation**:

$$\boxed{i\hbar\frac{\partial}{\partial t}|\psi\rangle = \hat{H}|\psi\rangle} \quad \text{and} \quad \boxed{P(a) = |\langle a|\psi|a|\psi\rangle|^2}$$

This course provides the common quantum mechanics toolkit that enables innovation across computer science, electrical engineering, computer engineering, and mechanical engineering applications.

1.6 A Note to Students

Initial Expectations for Lecture 1

This first lecture provides a **high-level overview** of quantum mechanics and its historical development. At this stage:

- **Don't worry if some concepts seem abstract or unfamiliar**—the meaning of mathematical symbols and physical ideas will become clearer as we progress through the course.
- **You may encounter unfamiliar notation** (Dirac notation, operators, Hilbert spaces)—these will be systematically introduced in subsequent lectures.
- **Some historical context may seem disconnected** from the modern formalism—we will show how these pieces fit together mathematically in later lectures.
- **The universal scope of QM** presented here is philosophical and conceptual; the practical mathematical tools for specific problems come later.

Think of this lecture as **laying the conceptual groundwork**—providing motivation and context before we dive into the mathematical details.

1.7 The Universal Scope of Quantum Mechanics

Quantum Mechanics as a Universal Framework

Quantum mechanics is a fundamental theoretical framework for physics that is considered valid across **all physical scales**:

Three Scales Where Quantum Mechanics Applies

Scale 1: Microscopic Scale (fundamental quantum realm):

- Atoms, molecules, subatomic particles (electrons, protons, neutrons)
- Elementary particles (quarks, leptons, gauge bosons)
- Quantum fields and vacuum fluctuations

Scale 2: Intermediate/Mesoscopic Scale (quantum-to-classical transition):

- Macroscopic quantum phenomena (Bose-Einstein condensates, superconductors)
- Nanoscale systems (quantum dots, molecular machines)
- Everyday objects where quantum effects are measurable (interference of large molecules)
- Biological quantum processes (photosynthesis, magnetoreception)

Scale 3: Cosmological Scale (quantum gravity and cosmology):

- Early universe physics (Big Bang, inflation)
- Quantum black holes and Hawking radiation
- Quantum cosmology and the wave function of the universe
- Quantum gravity (string theory, loop quantum gravity)

Classical Physics as an Emergent Phenomenon

The Quantum-to-Classical Transition

Classical physics emerges from quantum mechanics through several mechanisms:

- **Decoherence**: Environment-induced suppression of quantum interference
- **Classical limit** ($\hbar \to 0$): Formal mathematical limit
- **Ehrenfest's theorem**: Quantum expectation values follow classical equations

The precise nature of the quantum-to-classical transition remains an active research area in foundations of quantum mechanics.

Quantum Cosmology and the Wave Function of the Universe

Wheeler-DeWitt Equation and Quantum Cosmology

In quantum cosmology, the entire universe is described by a **wave function of the universe**:

$$\Psi_{\text{univ}}[\gamma_{ij}, \phi]$$

where γ_{ij} is the 3-metric of space and ϕ represents matter fields.

- Proposed by Bryce DeWitt and John Wheeler in the 1960s
- Stephen Hawking and James Hartle developed the "no-boundary proposal" in the 1980s
- Attempts to describe the quantum origin of spacetime itself
- The Wheeler-DeWitt equation: $\hat{H}\Psi_{\text{univ}} = 0$ (timeless formulation)

The Universal Reductionist Program

From Quantum Foundations to Complex Phenomena

The reductionist view suggests that all physical phenomena should ultimately be derivable from quantum mechanics:

$$\Psi_{\text{univ}} \xrightarrow{\text{quantum cosmology}} \text{fundamental physics} \xrightarrow{\text{emergence}} \begin{cases} \text{chemistry} \\ \text{biology} \\ \text{neuroscience} \\ \text{sociology} \\ \text{economics} \end{cases}$$

- This represents an ambitious (and controversial) research program
- Practical limitations: Exponential complexity makes direct derivation impossible
- Emergent phenomena often require their own effective theories
- The relationship between quantum foundations and higher-level phenomena remains an open philosophical question

Mathematical Hierarchy of Theories

$$\begin{array}{ccccc} \text{Quantum Gravity} & \supset & \text{Quantum Field Theory} & \supset & \text{Quantum Mechanics} \\ \cap & & \cap & & \cap \\ \text{General Relativity} & \supset & \text{Classical Field Theory} & \supset & \text{Classical Mechanics} \end{array}$$

Interpretation: The more fundamental theory (top row) contains the less fundamental theory (bottom row) as a special case or limiting approximation. For example:

- Quantum Field Theory $\supset$ Quantum Mechanics (in the non-relativistic limit)
- Quantum Mechanics $\supset$ Classical Mechanics (as $\hbar \to 0$)
- Quantum Gravity $\supset$ General Relativity (in the classical limit)

Each theory emerges as an approximation/limit of the more fundamental one above it

1.8 Further Reading

Core Textbooks

Feynman Lectures: The Enduring Classic

- **Complete Reference**: Richard P. Feynman, Robert B. Leighton, and Matthew Sands, *The Feynman Lectures on Physics, Vol. III: Quantum Mechanics*. Basic Books/Caltech, 1963.
- **Key Features**: Unique blend of wavefunction and Hilbert-space formalisms; unparalleled physical insights; includes advanced undergraduate topics like band structures and superconductors.
- **Accessibility**: Freely available online at https://www.feynmanlectures.caltech.edu/ with high-quality HTML presentation.

Susskind: The Modern Theoretical Minimum

- **Complete Reference**: Leonard Susskind and Art Friedman, *Quantum Mechanics: The Theoretical Minimum*. Basic Books/Penguin, 2014.
- **Key Features**: Firmly based on the modern Dirac-Hilbert space formalism while remaining accessible to undergraduates; includes modern topics like quantum entanglement; inspired by Feynman's pedagogical clarity.
- **Supplementary Materials**: Accompanied by video lectures available on YouTube.

Proposed Reading Methodology

Course Reading Strategy

- **Weekly Assignments**: I will assign specific sections from both books to complement lecture topics.
- **Feynman's Breadth**: Excellent for multidisciplinary audiences and unique applications.
- **Susskind's Modernity**: Provides the abstract mathematical framework essential for modern quantum information science.
- **Hybrid Approach**: Reading both provides both historical insight and modern formalism.

Foundational Texts: Historical and Conceptual

Classical Texts on Quantum Theory

- **David Bohm, *Quantum Theory*. Prentice-Hall, 1951; Dover Reprint.** A classic emphasizing physical intuition and conceptual foundations, with particularly clear discussions of the measurement problem.
- **P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed. Oxford University Press, 1958.** Written by one of QM's founders; most profitable after gaining initial familiarity with the subject. The definitive early statement of the Hilbert space formalism.
- **Roger Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*. Jonathan Cape, 2004.** Encyclopedic; quantum physics chapters scattered throughout offer exceptional insight into the mathematical structure of physical theory.

Deeper Second Books on Quantum Mechanics

Conceptual & Structural Texts on Quantum Theory

- **Chris J. Isham, *Lectures on Quantum Theory: Mathematical and Structural Foundations*. Imperial College Press, 1995.** A top-down, mathematically precise approach that builds quantum theory from its abstract Hilbert space foundations. Emphasizes structural clarity over historical narrative.
- **Jakob Schwichtenberg, *A No-Nonsense Introduction to Quantum Mechanics*. Independently published, 2020s.** Strips away historical digressions to present the logical and calculational core of QM in a clean, modern style. Ideal for building a practical understanding of the formalism while introducing deep and advanced topics.

Alternative Interpretations: Bohmian Mechanics

The Causal Interpretation (de Broglie-Bohm Theory)

- **Philosophical Foundation**: An alternative to mainstream interpretations that restores determinism and particle trajectories.
- **David Bohm and B. J. Hiley, *The Undivided Universe: An Ontological Interpretation of Quantum Theory*. Routledge, 1993.** The most complete statement of Bohm's mature interpretation.
- **Peter R. Holland, *The Quantum Theory of Motion: An Account of the de Broglie–Bohm Causal Interpretation*. Cambridge University Press, 1993.** A comprehensive textbook treatment.
- **Detlef Dürr and Stefan Teufel, *Bohmian Mechanics: The Physics and Mathematics of Quantum Theory*. Springer, 2009.** A modern, rigorous mathematical treatment.

Course Relevance: While time constraints prevent detailed coverage, I will reference this important framework throughout the course to highlight interpretive choices in standard QM.

Foundations and Advanced Undergraduate Perspectives

Foundations and Mathematical Approaches

- **Travis Norsen, *Foundations of Quantum Mechanics: An Exploration of the Physical Meaning of Quantum Theory.* Springer, 2017.** Recommended for undergraduates seeking a clear, modern treatment of interpretational issues.
- **Stephen Bruce Sontz, *An Introductory Path to Quantum Theory: Using Mathematics to Understand the Ideas of Physics.* Springer, 2020.** A mathematically rigorous yet accessible approach, emphasizing how mathematics structures physical understanding.

Physics in Broad Context

Quantum Mechanics in Scientific & Philosophical Context

- **Albert Einstein and Leopold Infeld, *The Evolution of Physics: From Early Concepts to Relativity and Quanta.* Simon & Schuster, 1938.** Situates QM within the historical development of physical concepts.
- **Werner Heisenberg, *Physics and Philosophy: The Revolution in Modern Science.* Harper & Row, 1958.** A founder's perspective on the philosophical implications.
- **Stephen Hawking, *A Brief History of Time: From the Big Bang to Black Holes.* Bantam Books, 1988.** The famous bestseller connecting quantum physics to cosmology.
- **Louis de Broglie, *The Revolution in Physics: A Non-Mathematical Survey of Quanta.* Noonday Press, 1953.** A conceptual overview by the originator of matter waves.

1.9 Mathematical Notation and Conventions

Basic Mathematical Objects

- **Scalar Fields**: $\mathbb{R}$ (real numbers), $\mathbb{C}$ (complex numbers)
- **Complex Unit**: $i = \sqrt{-1}$; complex conjugate: z^* or $\overline{z}$
- **Planck's Constant**: Reduced form $\hbar = h/(2\pi)$
- **Imaginary Unit in QM**: Always i, never j

Vectors and Spaces

- **Cartesian Vectors**: Boldface notation: $\mathbf{r}, \mathbf{p}, \mathbf{x}, \mathbf{q}$
- **Vector Components**: x, y, z or x_i for $i = 1, 2, 3$
- **Hilbert Space**: Abstract state space denoted $\mathcal{H}$
- **Function Space**: $L^2(\mathbb{R}^n) = \{\psi : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2 d^n x < \infty\}$

Quantum-Specific Notation

- **Dirac Notation**:
  - State vector (ket): $|\psi\rangle \in \mathcal{H}$
  - Dual vector (bra): $\langle\phi| \in \mathcal{H}^*$
  - Inner product: $\langle\phi|\psi\rangle$
- **Operators**:
  - Hat notation: $\hat{A}, \hat{x}, \hat{p}, \hat{H}$
  - Hermitian conjugate: $\hat{A}^\dagger$
- **Commutator**: $[\hat{A}, \hat{B}] := \hat{A}\hat{B} - \hat{B}\hat{A}$
- **Position Basis**: Wavefunction: $\psi(\mathbf{r}) = \langle\mathbf{r}|\psi\rangle$

1.10 Important Constants

Constant	Symbol	Value
Planck's constant	h	6.626×10^{-34} J·s
Reduced Planck constant	$\hbar = h/2\pi$	1.055×10^{-34} J·s
Speed of light	c	2.998×10^{8} m/s
Electron mass	m_e	9.109×10^{-31} kg
Proton mass	m_p	1.673×10^{-27} kg
Bohr radius	a_0	5.292×10^{-11} m
Rydberg constant	R_H	1.097×10^{7} m^{-1}

2 Review of Math and Physics Background

2.1 Review of Complex Numbers

The imaginary unit i is defined as a number satisfying $i^2 = -1$. Formally, it can be constructed within set theory as the ordered pair $(0, 1)$, which in turn can be defined as $\{\{0\}, \{0, 1\}\}$. This construction shows that complex numbers exist within rigorous mathematics. Quantum mechanics has demonstrated that the structure of the physical world itself relies fundamentally on complex numbers.

Definition 1 (Complex Number). *A complex number z can be written in two equivalent forms:*

- ***Rectangular form**: $z = x + iy$, where $x, y \in \mathbb{R}$*
- ***Polar form**: $z = re^{i\theta}$, where $r \geq 0$ and $\theta \in [0, 2\pi)$*

Here $r = |z| = \sqrt{x^2 + y^2}$ is the magnitude *and $\theta = \arg(z)$ is the* phase*.*

Definition 2 (Complex Conjugate). *For $z = x + iy$, the complex conjugate is $z^* = x - iy$. In polar form, if $z = re^{i\theta}$, then $z^* = re^{-i\theta}$.*

Euler's Identity (Fundamental Relation)

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

This remarkable identity connects exponential, trigonometric, and complex number theory. Two special cases are particularly important:

$$e^{i\pi} + 1 = 0 \quad \text{and} \quad e^{i\pi/2} = i$$

2.2 Review of Vector and Hilbert Spaces

We provide only the minimum needed to define Hilbert spaces. Think of a Hilbert space as a generalization of the familiar Euclidean vector space $\mathbb{R}^3$.

Example 1 (Euclidean Space $\mathbb{R}^3$). *The space $\mathbb{R}^3$ consists of all vectors $\mathbf{r}$ that can be written as:*

$$\mathbf{r} = \mathbf{i}r_1 + \mathbf{j}r_2 + \mathbf{k}r_3$$

where $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ are three orthonormal basis vectors.

From this example, we extract three key concepts:

1. **Normalization**: $\|\mathbf{i}\| = \|\mathbf{j}\| = \|\mathbf{k}\| = 1$
2. **Orthonormality**: $\mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0$
3. **Completeness**: Any vector $\mathbf{r} \in \mathbb{R}^3$ can be expressed as a linear combination of the basis vectors

Definition 3 (Hilbert Space). *A Hilbert space $\mathcal{H}$ is a complete inner product space over the complex numbers $\mathbb{C}$.*

- **Vectors**: In Dirac notation, a vector in $\mathcal{H}$ is written as a *ket*: $|\psi\rangle$
- **Basis Expansion**: An arbitrary ket can be expanded in a (possibly countably infinite) basis $\{|\phi_n\rangle\}$:

$$|\psi\rangle = \sum_n c_n |\phi_n\rangle, \quad c_n \in \mathbb{C}$$

- **Dual Vectors**: The dual of $|\psi\rangle$ is written as a *bra*: $\langle\psi|$. If we think of $|\psi\rangle$ as a column vector, then $\langle\psi|$ is its conjugate transpose:
$$\langle\psi| = |\psi\rangle^\dagger$$
- **Inner Product**: For any two kets $|\phi\rangle, |\psi\rangle \in \mathcal{H}$, the inner product is written as $\langle\phi|\psi\rangle \in \mathbb{C}$.
- **Norm**: The length (norm) of $|\psi\rangle$ is $\| \, |\psi\rangle \, \| = \sqrt{\langle\psi|\psi\rangle}$.

Definition 4 (Operator)**.** *An operator $\hat{A}$ is a linear map $\hat{A} : \mathcal{H} \to \mathcal{H}$.*

Example 2. *In $\mathbb{R}^3$, operators are 3×3 matrices. In Hilbert spaces, operators can be infinite-dimensional matrices.*

Definition 5 (Adjoint Operator)**.** *For an operator $\hat{A}$, its adjoint $\hat{A}^\dagger$ is defined by the relation*

$$\langle\psi|\hat{A}^\dagger|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$

for all vectors $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, where * denotes complex conjugation. Equivalently,

$$\langle\psi|\hat{A}^\dagger\phi\rangle = \langle\hat{A}\psi|\phi\rangle$$

for all $|\psi\rangle, |\phi\rangle$.

Definition 6 (Algebraic Properties of the Adjoint)**.** *The adjoint operation satisfies the following properties for all operators $\hat{A}$, $\hat{B}$ and scalars $\alpha \in \mathbb{C}$:*

(i) ***Adjoint of an adjoint:*** $(\hat{A}^\dagger)^\dagger = \hat{A}$.

(ii) ***Adjoint of a sum:*** $(\hat{A} + \hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$.

(iii) ***Adjoint of a product:*** $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ *(reverse order).*

(iv) ***Adjoint of a scalar multiple:*** $(\alpha\hat{A})^\dagger = \alpha^*\hat{A}^\dagger$.

(v) ***Adjoint of the identity:*** $\hat{I}^\dagger = \hat{I}$.

Remark 2.1 (Connection to Matrix Mechanics)**.** *In a finite-dimensional Hilbert space with an orthonormal basis, if $\hat{A}$ is represented by the matrix $\mathbf{A}$, then $\hat{A}^\dagger$ is represented by the conjugate transpose matrix $\mathbf{A}^\dagger = (\mathbf{A}^T)^*$. The product rule $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$ corresponds to the matrix identity $(\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger$.*

Theorem 1 (Eigenvector Relation for $\hat{A}^\dagger$)**.** *Let $\hat{A}$ be an operator on a Hilbert space $\mathcal{H}$ with an eigenvector $|a\rangle$ and corresponding eigenvalue λ, i.e. $\hat{A}|a\rangle = \lambda|a\rangle$. Then the adjoint operator $\hat{A}^\dagger$ satisfies the following:*

$$\langle a|\hat{A}^\dagger = \lambda^*\langle a|$$

That is, $\langle a|$ is a ***left eigenvector*** *of $\hat{A}^\dagger$ with eigenvalue λ^*.*

However, $|a\rangle$ is an eigenvector of $\hat{A}^\dagger$ ***if and only if*** *$\hat{A}$ is a normal operator ($[\hat{A}, \hat{A}^\dagger] = 0$). In that case:*

$$\hat{A}^\dagger|a\rangle = \lambda^*|a\rangle$$

Definition 7 (Hermitian (Self-Adjoint) Operator)**.** *An operator $\hat{A}$ is Hermitian (or self-adjoint) if $\hat{A}^\dagger = \hat{A}$. In terms of inner products, this means*

$$\langle\psi|\hat{A}|\phi\rangle = \langle\phi|\hat{A}|\psi\rangle^*$$

for all $|\psi\rangle, |\phi\rangle \in \mathcal{H}$, or equivalently,

$$\langle\hat{A}\psi|\phi\rangle = \langle\psi|\hat{A}\phi\rangle.$$

Definition 8 (Eigenvalue Problem)**.** *For an operator $\hat{A}$, the eigenvalue equation is:*

$$\hat{A}\,|\varphi_n\rangle = \lambda_n\,|\varphi_n\rangle$$

where λ_n are eigenvalues and $|\varphi_n\rangle$ are eigenvectors.

Theorem 2 (Properties of Hermitian Operators)**.** *For a Hermitian operator $\hat{A}$:*

1. *All eigenvalues are real: $\lambda_n \in \mathbb{R}$*
2. *Eigenvectors corresponding to distinct eigenvalues are orthogonal*
3. *The eigenvectors form a complete orthonormal basis for $\mathcal{H}$ (provided certain technical conditions)*

The Space of Square-Summable Sequences

One of the simplest infinite-dimensional Hilbert spaces is the space ℓ^2 of square-summable sequences:

$$\ell^2 = \left\{ (c_1, c_2, c_3, \dots) \in \mathbb{C}^\infty : \sum_{n=1}^{\infty} |c_n|^2 < \infty \right\}.$$

The inner product is defined as:

$$\langle a|b\rangle = \sum_{n=1}^{\infty} a_n^* b_n.$$

The condition $\sum_{n=1}^{\infty} |c_n|^2 < \infty$ ensures that the vectors have finite norm (often interpreted as finite "energy" in physical contexts). This is the natural infinite-dimensional generalization of $\mathbb{C}^n$.

The Space of Square-Integrable Functions

The most important Hilbert space for quantum mechanics is the space $L^2(\mathbb{R}^n)$ of square-integrable functions:

$$L^2(\mathbb{R}^n) = \left\{ \psi : \mathbb{R}^n \to \mathbb{C} : \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty \right\}.$$

The inner product is:

$$\langle \phi|\psi\rangle = \int_{\mathbb{R}^n} \phi^*(\mathbf{x})\,\psi(\mathbf{x})\, d^n x.$$

Wavefunctions $\psi(\mathbf{r}, t)$ in non-relativistic quantum mechanics are elements of $L^2(\mathbb{R}^3)$ for a single particle, or $L^2(\mathbb{R}^{3N})$ for N particles.

The Relation Between ℓ^2 and L^2

Given an orthonormal basis $\{\varphi_n(\mathbf{x})\}$ of $L^2(\mathbb{R}^n)$, any wavefunction $\psi(\mathbf{x})$ can be expanded as:

$$\psi(\mathbf{x}) = \sum_{n=1}^{\infty} c_n \varphi_n(\mathbf{x}),$$

where the coefficients $c_n = \langle \varphi_n|\psi\rangle$ satisfy:

$$\sum_{n=1}^{\infty} |c_n|^2 = \int_{\mathbb{R}^n} |\psi(\mathbf{x})|^2\, d^n x < \infty.$$

This establishes an isomorphism between $L^2(\mathbb{R}^n)$ and ℓ^2: the square-integrable function ψ corresponds uniquely to the square-summable sequence $\{c_n\}$ of its expansion coefficients.

Orthonormal Vectors

A set of vectors $\{|\phi_n\rangle\}$ in a Hilbert space $\mathcal{H}$ is called **orthonormal** if it satisfies:

$$\langle \phi_m|\phi_n\rangle = \delta_{mn} = \begin{cases} 1 & \text{if } m = n, \\ 0 & \text{if } m \neq n. \end{cases}$$

Equivalently:

- **Orthogonality**: $\langle \phi_m|\phi_n\rangle = 0$ for $m \neq n$ (vectors are perpendicular)
- **Normalization**: $\| \, |\phi_n\rangle \, \| = \sqrt{\langle \phi_n|\phi_n\rangle} = 1$ (each vector has unit length)

An orthonormal set that is also **complete** (i.e., spans the entire space $\mathcal{H}$) forms an **orthonormal basis**. Any vector $|\psi\rangle \in \mathcal{H}$ can then be expanded as $|\psi\rangle = \sum_n c_n |\phi_n\rangle$ with $c_n = \langle \phi_n|\psi\rangle$.

Here's your revised subsection with a review of statistical independence added:

2.3 Review of Basic Probability Theory

We need only the minimal foundations of probability theory.

Definition 9 (Probability Space)**.** *A probability space consists of:*

- ***Sample space*** Ω*: The set of all possible outcomes*
- ***Events****: A* σ*-algebra* $\mathcal{F}$ *of subsets of* Ω
- ***Probability measure*** $\mathbb{P}$*: A function* $\mathbb{P} : \mathcal{F} \to [0, 1]$ *satisfying:*
  1. $\mathbb{P}(\Omega) = 1$
  2. *Countable additivity: For disjoint events* A_i*,* $\mathbb{P}(\bigcup_i A_i) = \sum_i \mathbb{P}(A_i)$

Definition 10 (Independence of Events)**.** *Two events* A *and* B *are* ***independent*** *if*

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Equivalently, assuming $\mathbb{P}(B) > 0$*,* A *and* B *are independent if* $\mathbb{P}(A \mid B) = \mathbb{P}(A)$*.*

Definition 11 (Conditional Probability)**.** *For events* A *and* B *with* $\mathbb{P}(B) > 0$*, the* ***conditional probability*** *of* A *given* B *is*

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The concept of independence generalizes to collections of events. Events $A_1, A_2, \ldots, A_n$ are **mutually independent** if for every subset of indices $I \subseteq \{1, \ldots, n\}$,

$$\mathbb{P}\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} \mathbb{P}(A_i).$$

2.4 Review of Classical Mechanics

2.4.1 The Lagrangian Formalism: A Brief Review

The Lagrangian formulation provides a powerful and elegant framework for classical mechanics, operating in **configuration space** rather than phase space.

The Configuration Space Description

For a system of N particles in three-dimensional space, the configuration is specified by $3N$ generalized coordinates. These can be taken as the Cartesian coordinates of each particle:

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_N) \in \mathbb{R}^{3N},$$

where each $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ is the position vector of particle α. The full state at a given time also includes the velocities $\dot{\mathbf{q}} = (\dot{\mathbf{r}}_1, \dot{\mathbf{r}}_2, \ldots, \dot{\mathbf{r}}_N)$, giving a point in the **tangent bundle** of the configuration space $\mathbb{R}^{3N}$.

The dynamics are governed by the **Lagrangian function**, a map from the tangent bundle of configuration space to the real numbers:

$$L : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \dot{\mathbf{q}}) \mapsto L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

which typically represents the difference between kinetic and potential energy.

The time evolution follows from **Hamilton's principle of stationary action**:

$$\delta S = \delta \int_{t_1}^{t_2} L(\mathbf{q}, \dot{\mathbf{q}}, t)\, dt = 0, \tag{1}$$

which yields the **Euler-Lagrange equations** for each degree of freedom:

$$\boxed{\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0 \qquad (i = 1, \ldots, 3N)}. \tag{2}$$

- **Standard form for N particles**: The Lagrangian is typically the difference between total kinetic and potential energy:

  $$L(\mathbf{r}_1, \ldots, \mathbf{r}_N, \dot{\mathbf{r}}_1, \ldots, \dot{\mathbf{r}}_N) = \sum_{\alpha=1}^{N} \frac{1}{2} m_\alpha \dot{\mathbf{r}}_\alpha^2 - V(\mathbf{r}_1, \ldots, \mathbf{r}_N).$$

  The Euler-Lagrange equations for coordinate x_α then give $m_\alpha \ddot{x}_\alpha = -\partial V / \partial x_\alpha$, recovering Newton's second law.

- **Canonical momentum**: The momentum conjugate to the coordinate q_i (which could be, e.g., x_α) is:

  $$p_i := \frac{\partial L}{\partial \dot{q}_i}.$$

  For the Cartesian coordinates above, $\mathbf{p}_\alpha = m_\alpha \dot{\mathbf{r}}_\alpha$.

- **Legendre transformation to Hamiltonian**: The Hamiltonian is obtained via:

  $$H(\mathbf{q}, \mathbf{p}, t) = \sum_{i=1}^{3N} p_i \dot{q}_i - L(\mathbf{q}, \dot{\mathbf{q}}, t),$$

  where the velocities $\dot{q}_i$ are expressed in terms of $(\mathbf{q}, \mathbf{p})$ using the momentum definition. For the standard particle Lagrangian, this yields:

  $$H = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \ldots, \mathbf{r}_N).$$

This formulation—based on a single scalar function L and the principle of stationary action—provides a coordinate-independent foundation for mechanics and directly motivates the path integral approach to quantization.

2.4.2 The Classical Hamiltonian Formalism: A Review

Before examining how quantum mechanics modifies classical mechanics, it is essential to recall the core structure of the Hamiltonian formulation of classical dynamics.

The Phase Space Description

For a system of N particles in three-dimensional space, the complete state is specified by $3N$ generalized position coordinates and their $3N$ conjugate momenta. Using Cartesian coordinates, we write:

$$\mathbf{q} = (\mathbf{r}_1, \mathbf{r}_2, \dots, \mathbf{r}_N) \in \mathbb{R}^{3N}, \quad \mathbf{p} = (\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_N) \in \mathbb{R}^{3N},$$

where $\mathbf{r}_\alpha = (x_\alpha, y_\alpha, z_\alpha)$ is the position of particle α and $\mathbf{p}_\alpha = (p_{\alpha x}, p_{\alpha y}, p_{\alpha z})$ is its conjugate momentum. The instantaneous state is a point $(\mathbf{q}, \mathbf{p})$ in a $6N$-dimensional **phase space** $\mathbb{R}^{6N}$.

The dynamics are governed by the **Hamiltonian function**, a map from phase space to the real numbers:

$$H : \mathbb{R}^{6N} \to \mathbb{R}, \quad (\mathbf{q}, \mathbf{p}) \mapsto H(\mathbf{q}, \mathbf{p}),$$

which typically represents the total energy of the system. The time evolution is given by **Hamilton's canonical equations**:

$$\boxed{\dot{q}_i = \frac{\partial H}{\partial p_i}, \qquad \dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad (i = 1, \dots, 3N)}. \tag{3}$$

- **Standard form for N particles**: For a system with conservative forces, the Hamiltonian is:

$$H(\mathbf{r}_1, \dots, \mathbf{r}_N, \mathbf{p}_1, \dots, \mathbf{p}_N) = \sum_{\alpha=1}^{N} \frac{\mathbf{p}_\alpha^2}{2m_\alpha} + V(\mathbf{r}_1, \dots, \mathbf{r}_N).$$

Hamilton's equations then yield Newton's second law for each particle. For particle α:

$$\dot{\mathbf{r}}_\alpha = \frac{\partial H}{\partial \mathbf{p}_\alpha} = \frac{\mathbf{p}_\alpha}{m_\alpha}, \quad \dot{\mathbf{p}}_\alpha = -\frac{\partial H}{\partial \mathbf{r}_\alpha} = -\nabla_{\mathbf{r}_\alpha} V \quad \Rightarrow \quad m_\alpha \ddot{\mathbf{r}}_\alpha = -\nabla_{\mathbf{r}_\alpha} V.$$

This elegant structure in phase space—with its clear separation of coordinates and momenta—provides the direct mathematical foundation for the **canonical quantization procedure** that follows.

3 What is Quantum Mechanics: Two Approaches to the Same Topic

3.1 Two Distinct Approaches to Quantum Mechanics

Remarkable Duality in Understanding Quantum Theory

Quantum mechanics can be approached through **two fundamentally different perspectives**: one emphasizing physical intuition and empirical foundations, the other emphasizing abstract mathematical structure and algorithmic principles.

Physical Space Approach (Quantization)
Historically first, physics-driven, not unified

↓ Historical development

Abstract/Algorithmic Approach
Modern, mathematics-driven, structurally clean

The Physical Space Approach (Quantization)

From Classical to Quantum

This approach, historically the first to be developed, emphasizes the **physical content** of quantum mechanics through the process of *quantization*.

- **Historical precedence**: Developed during the quantum revolution (1900-1927)
- **Methodology**: Classical system $\xrightarrow{\text{quantization}}$ Quantum system
- **Core philosophy**: Quantum mechanics as *modification* of classical physics
- **Diverse techniques**: Not a single method but a collection of approaches:
  - Canonical quantization: $q \to \hat{q}$, $p \to \hat{p}$, $[\hat{q}, \hat{p}] = i\hbar$
  - Path integral quantization: Sum over histories (Feynman)
  - Geometric quantization: Symplectic geometry methods
  - Deformation quantization: Star product formalism
  - Stochastic quantization: Random process methods

Physical Approach Flow: $\boxed{\text{Classical Physics}} \xrightarrow[\text{quantization}]{} \boxed{\text{Quantum Physics}}$

The Abstract/Algorithmic Approach

Quantum as Information Processing Framework

This modern approach emphasizes the **mathematical structure** of quantum mechanics, treating it as an abstract computational or information-theoretic framework. It has become dominant in quantum information theory and computer science.

- **Historical emergence**: Gained prominence in late 20th century with quantum information science
- **Methodology**: Postulates → Mathematical structure → Applications
- **Key insight**: Quantum mechanics as *generalized probability theory*
- **Revolutionary feature**: Can formulate entire quantum theory **without ever mentioning the Schrödinger equation for specific physical systems**
- **Main mathematical tools**:
  - **Linear algebra**: Hilbert spaces, eigenvectors, spectral theorem
  - **Operator theory**: C^*-algebras, von Neumann algebras, completely positive maps
  - **Information theory**: Entropy measures, channel capacities, resource theories
  - **Category theory**: Monoidal categories, string diagrams

  Abstract Approach Flow: [Mathematical Postulates] → [Algebraic Structure] → [Physical/Computational Applications]

Comparison of Approaches

Aspect	Physical (Quantization)	Abstract/Algorithmic
Starting point	Classical physics	Mathematical postulates
Emphasis	Physical interpretation	Structural properties
Historical period	1900-1927 (development)	1930-present (refinement)
Dominant in	Traditional physics education	Quantum information, computer science
Key equation	Schrödinger equation	Postulates + linear algebra
Main tools	Differential equations, operators	Linear algebra, operator theory, information theory

Pedagogical Strategy for This Course

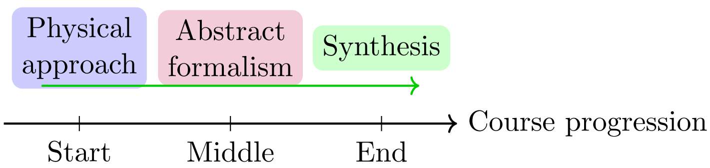
Hybrid Approach: Best of Both Worlds

We adopt a **hybrid pedagogical strategy** that combines the strengths of both approaches:

Phase 1: Build physical intuition: Start with wave mechanics and quantization

Phase 2: Develop abstract formalism: Introduce Hilbert space, Dirac notation, operator methods

Phase 3: Synthesize approaches: Show equivalence and complementary insights



Why This Hybrid Approach?

Balancing Intuition and Rigor

The physical approach provides **concrete intuition** while the abstract approach provides **mathematical clarity**. Together they offer a complete understanding.

Advantages of starting physical:

- Connects to experimental physics students already know
- Provides concrete problem-solving methods
- Historical development mirrors conceptual understanding

Advantages of abstract formalism:

- Reveals logical structure independent of specific systems
- Essential for quantum information and computation
- Clean mathematical framework for advanced topics

3.2 The First Worldview: Quantization, From Classical to Quantum

The Physical Space Approach Foundation

To understand the physical space approach to quantum mechanics, we begin with classical mechanics concepts. Quantum theory in this view emerges through quantization of classical systems.

Three Types of Physical Space in Classical Mechanics

In classical mechanics, we distinguish three distinct but related spaces:

Space 1: Physical Spacetime:

- For non-relativistic quantum mechanics: $\mathbb{R}^3 \times \mathbb{R} \cong \mathbb{R}^4$
- Represents actual physical space (x, y, z) and time t
- Where particles and fields physically exist

Space 2: Configuration Space:

- Used in Lagrangian mechanics
- For N particles: $\mathbb{R}^{3N}$ degrees of freedom
- Generalized coordinates: $\mathbf{q}_i$, $i \in \{1, 2, \ldots, N\}$
- Describes all possible positions simultaneously

Space 3: Phase Space:

- Used in Hamiltonian mechanics
- For N particles: $\mathbb{R}^{6N}$ dimensions
- Generalized positions $\mathbf{q}_i$ and conjugate momenta $\mathbf{p}_i$
- Describes both positions and momenta simultaneously

Physical Space	Configuration Space	Phase Space
$\mathbb{R}^3 \times \mathbb{R}$	$\mathbb{R}^{3N}$	$\mathbb{R}^{6N}$
(x, y, z, t)	$\{\mathbf{q}_1, \ldots, \mathbf{q}_N\}$	$\{(\mathbf{q}_i, \mathbf{p}_i)\}$

The Hamiltonian Foundation of Quantum Mechanics

Hamiltonian Formalism as the Bridge

Conventionally, quantum mechanics in the physical space approach employs the Hamiltonian formalism. A key feature of this worldview is that every quantum system is thought to arise from quantizing some classical system.

- **Philosophical foundation**: Quantum mechanics as *modified* classical mechanics
- **Even for non-classical observables**: When quantum observables lack classical analogues (e.g., spin), one often inserts corresponding terms in the classical Hamiltonian
- **Methodology**: Builds on classical intuition, then introduces fundamentally non-classical features
- **Wave function location**: Despite using Hamiltonian formalism, the wave function ψ lives in configuration space, not phase space

Transition to Wave Mechanics: The Wavefunction as the State

The Radical Change in the State Concept

The physical space approach introduces a fundamental shift: the quantum state replaces the classical phase-space point. This state is not a set of numbers but a complex-valued field—the wavefunction $\psi(\mathbf{r}, t)$.

- **Classical State Space**: For a single particle, the instantaneous state is $(\mathbf{x}, \dot{\mathbf{x}})$ or $(\mathbf{q}, \mathbf{p})$, living in $\mathcal{S} \cong \mathbb{R}^6$. Observables are functions on $\mathcal{S}$.
- **Quantum State Space**: The state is $\psi(\mathbf{r}, t) \in L^2(\mathbb{R}^3)$, a function in configuration space. The collection of all such functions forms a Hilbert space $\mathcal{H}$—a much richer mathematical structure.
- **Information Content**: While a classical state specifies exact values, $\psi(\mathbf{r}, t)$ encodes probabilistic information about all possible measurements.

Physical Interpretation: Born's Rule

From Field to Probability

The wavefunction's physical meaning is provided by the Born rule, which connects the complex field to measurable probabilities.

$$\boxed{P(\text{particle in volume } V) = \int_V |\psi(\mathbf{r}, t)|^2 \, d^3\mathbf{r}}$$

- **Probability Density**: $\rho(\mathbf{r}, t) := |\psi(\mathbf{r}, t)|^2$ acts as a probability density function.
- **Normalization**: States are normalized: $\int_{\mathbb{R}^3} |\psi|^2 \, d^3\mathbf{r} = 1$, ensuring total probability equals 1.
- **Inherent Randomness**: Unlike classical mechanics, QM predictions are fundamentally probabilistic; the wavefunction embodies this indeterminacy.

Contrast with Classical Fields

Wavefunction vs. Physical Fields

The wavefunction is *not* analogous to classical fields like the electromagnetic field—it represents a deeper conceptual shift.

Quantum Mechanics (Wavefunction)	Classical Field Theory (e.g., EM)
ψ is the physical state of the particle	Fields are produced by charges/currents
Particle $\Longleftrightarrow$ Wavefunction (dual nature)	Source $\longrightarrow$ Field (causal relationship)
Born rule gives probabilistic interpretation	Field strength gives force/energy density

- **No Source Problem**: In standard QM, ψ has no known "source" equation like $\nabla \cdot \mathbf{E} = \rho/\epsilon_0$; it is fundamental.
- **Quantum Field Theory Perspective**: In QFT, this distinction blurs—fields become operator-valued and particles are excitations. Yet in non-relativistic QM, the wavefunction remains the primary description.

Classical Mechanics: States as Measurable Quantities

- The state $(\mathbf{q}, \mathbf{p}) \in \mathcal{S}$ (a phase space point) is directly measurable.
- Observables are functions $f : \mathcal{S} \to \mathbb{R}$; choosing the identity function $f_{\text{id}}(\mathbf{q}, \mathbf{p}) = (\mathbf{q}, \mathbf{p})$ gives the state itself. *Thus in classical physics, states in themselves are directly measurable. This is not the case in quantum physics.*
- State space $\mathcal{S}$ and observable space are essentially the same mathematical object.

Quantum Mechanics: The Radical Separation

- States $\psi \in \mathcal{H}$ (wavefunctions) are not directly measurable.
- Observables are operators $\hat{A} : \mathcal{H} \to \mathcal{H}$ (linear maps from states to states).
- States are abstract vectors; observables are defined independently as Hermitian operators.
- The relationship between state and measurement outcome is probabilistic:

$$P(a|\psi) = \langle \psi | P_a | \psi \rangle$$

  where a is an eigenvalue of $\hat{A}$ and P_a is the projection onto the corresponding eigenspace.

Philosophical Implication: The Abstract Construction of QM

- In QM, one must first construct abstract states (the Hilbert space $\mathcal{H}$), then define operators representing observables, and only then combine them via the Born rule to compute probabilities.
- The state-observable relationship becomes mediated by mathematical formalism rather than direct measurement.
- This represents a fundamental epistemological shift: knowledge of a quantum system is inherently inferential and probabilistic, derived from the interaction between abstract states and operator-valued observables.

The Quantization Procedure

From Classical to Quantum: The Quantization Recipe

The core of the physical space approach is the quantization procedure that transforms classical systems into quantum ones.

Step 1: Start with classical Hamiltonian:

$$H(\mathbf{q}, \mathbf{p}) = \frac{\mathbf{p}^2}{2m} + V(\mathbf{q})$$

Step 2: Promote variables to operators (canonical quantization):

$$\mathbf{q} \to \hat{\mathbf{q}}, \quad \mathbf{p} \to \hat{\mathbf{p}}, \quad \text{with} \quad [\hat{q}_i, \hat{p}_j] = i\hbar\delta_{ij}$$

Step 3: Construct quantum observables:

- Position: $\hat{x}$ (multiplication operator)
- Momentum: $\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$
- Angular momentum: $\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$
- Hamiltonian: $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{q}})$

Step 4: Introduce wave function:

$$\psi(\mathbf{q}, t) \in L^2(\mathbb{R}^{3N})$$

defined on configuration space despite Hamiltonian formulation

Quantization Flow:

$$\boxed{\text{Classical } H(\mathbf{q}, \mathbf{p})}$$
$$\downarrow \text{quantization}$$
$$\boxed{\text{Quantum } \hat{H}}$$
$$\downarrow$$
$$\boxed{\text{Schrödinger Equation}}$$
$$\downarrow$$
$$\boxed{\text{Wave function } \psi(\mathbf{q}, t)}$$

Beyond Classical Intuition: Intrinsic Quantum Properties

Where Classical Intuition Fails

Despite its classical foundation, quantum mechanics introduces phenomena with no classical counterparts:

- **Spin**: Intrinsic angular momentum (not orbital motion)
- **Isospin**: Internal symmetry of strong interactions
- **Parity**: Discrete spatial inversion symmetry
- **Quantum entanglement**: Non-classical correlations violating Bell inequalities
- **Superposition**: States existing in multiple configurations simultaneously

Physical Space Approach: Strengths and Limitations

- **Strengths**:
  - Builds directly on familiar classical concepts
  - Provides concrete calculation methods for specific systems
  - Historical development mirrors conceptual understanding
  - Natural for problems with clear classical limits
- **Quantum Surprises**:
  - Relevant functional space: Wave function lives in configuration space, not physical space
  - Measurement problem: How/when does quantum $\to$ classical transition occur?
  - Wave function collapse: How/when does ψ collapse?
  - Non-locality: Entanglement suggests connections beyond spacetime
  - Conceptual tension: Quantum systems from classical roots, yet fundamentally different

3.3 The Second Worldview: A Compact Résumé of the Quantum Algorithm

In this section we provide the minimum content of quantum mechanics condensed into four fundamental postulates. This can be done most elegantly using the Dirac-Hilbert space formalism. We will only review below the minimum mathematics required to understand the postulates. A more detailed exposition of the Dirac-Hilbert formulation will be given in later lectures.

Pedagogical Note

This section presents quantum mechanics in its *most abstract form*. Do not be alarmed if some concepts seem unfamiliar—we will build physical intuition gradually. Think of this as the "bare skeleton" of quantum theory; we will add the "flesh" of physical interpretation throughout the course.

3.3.1 Postulates of Quantum Mechanics

The Four-Postulate Algorithm

These four postulates provide a complete recipe for describing quantum systems. Remarkably, they contain no explicit mention of waves, particles, or the Schrödinger equation—these will emerge as consequences.

Postulate I: The Quantum State

Postulate 1 (State Postulate)**.** *Each quantum system S is associated with a Hilbert space $\mathcal{H}_S$. The instantaneous state of the system at time t is completely described by a vector $|\psi(t)\rangle \in \mathcal{H}_S$ with unit norm:*

$$\boxed{\langle\psi(t)|\psi(t)\rangle = 1}$$

Remark 3.1. *The state vector $|\psi(t)\rangle$ contains* all possible information *about the system. Different physical situations correspond to different Hilbert spaces (e.g., spin-*$1/2$ *system: $\mathbb{C}^2$, particle in 1D: $L^2(\mathbb{R})$).*

Postulate II: The Operators

Postulate 2 (Operator Postulate)**.** *There are two fundamental types of operators in quantum mechanics:*

(a) ***Observables (Measurement Operators)****: Each physically measurable quantity is represented by a unique Hermitian operator $\hat{A}$. The possible measurement outcomes are precisely the real eigenvalues λ_n of $\hat{A}$.*

(b) ***Evolution Operator****: For a closed quantum system S, the time evolution from initial state $|\psi(t_0)\rangle$ to final state $|\psi(t)\rangle$ is described by a unitary operator $\hat{U}(t, t_0)$:*

$$\boxed{|\psi(t)\rangle = \hat{U}(t, t_0)\,|\psi(t_0)\rangle}$$

where $\hat{U}^\dagger \hat{U} = \hat{U}\hat{U}^\dagger = \hat{I}$ (unitarity).

Postulate III: Measurement and Collapse

Postulate 3 (Measurement Postulate - Born Rule)**.** *When measuring an observable $\hat{A}$ on a system in state $|\psi\rangle$, the following occurs:*

1. ***Outcome****: The measurement yields one of the eigenvalues λ_n of $\hat{A}$*

2. ***Probability****: The probability of obtaining λ_n is*

$$\boxed{P(\lambda_n) = |\,\langle \varphi_n|\psi\rangle\,|^2}$$

where $|\varphi_n\rangle$ is the eigenvector corresponding to λ_n

3. ***Collapse****: Immediately after measurement, the state collapses to the corresponding eigenstate:*

$$\boxed{|\psi\rangle \xrightarrow{\text{measure } \lambda_n} \frac{|\varphi_n\rangle}{\|\,|\varphi_n\rangle\,\|}}$$

Remark 3.2. *This is the most controversial postulate—it introduces true randomness and discontinuous state collapse. The measurement process itself is not described by unitary evolution.*

Postulate IV: Composite Systems

Postulate 4 (Composite Systems Postulate)**.** *If system S consists of two subsystems S_1 and S_2 with Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, then the Hilbert space of the composite system is the tensor product:*

$$\boxed{\mathcal{H}_S = \mathcal{H}_1 \otimes \mathcal{H}_2}$$

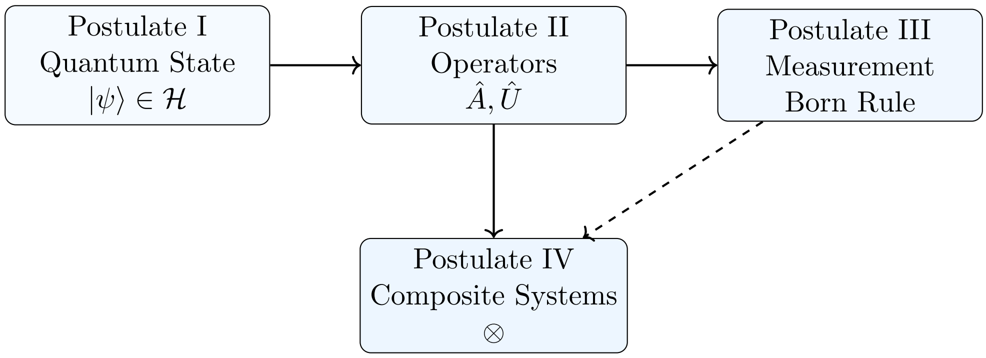
If $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$, then the composite state is $|\psi_1\rangle \otimes |\psi_2\rangle \in \mathcal{H}_1 \otimes \mathcal{H}_2$.

Remark 3.3 (Tensor Product)**.** *We will define the tensor product rigorously in later lectures. For now, think of it as a way to combine systems while preserving the individuality of each subsystem. This postulate enables quantum entanglement—states that cannot be written as simple products $|\psi_1\rangle \otimes |\psi_2\rangle$.*

3.3.2 Completeness and Emergent Concepts

We may now summarize the quantum algorithm using the following diagrammatic structure:

The Quantum Algorithm



- **Postulate I (The Stage):** The existence of the quantum state and Hilbert space creates the fundamental mathematical “stage” upon which the entire theory is built. Within this structured space, we define **operators**, which encode virtually all physical information about a system—both what can be observed in principle (through Hermitian observables) and how it evolves in time (through unitary evolution operators).

- **The Primacy of Operators:** Some advanced formulations, such as *algebraic quantum mechanics*, highlight the deep foundational role of operators. They demonstrate that one can even dispense with the explicit concept of a quantum state vector and work entirely within the algebra of operators, underscoring their centrality.

- **Postulate II (Dynamics and Observables):** This postulate sets the stage for extracting physical information. It defines what can be measured (observables) and dictates the deterministic evolution of the state (unitary operators), thereby preparing the system for the critical act of **measurement**.

- **Postulate III (Measurement and Collapse):** Governs the actual extraction of information. Through the Born rule, it bridges the abstract formalism with empirical reality by providing probabilities for measurement outcomes. It also describes the state’s **discontinuous collapse** into an eigenstate upon observation, completing the connection between theory and experiment.

- **Postulate IV (Building Complexity):** Serves as the architectural or recursive rule for constructing complex systems. It describes how individual subsystems, each with their own Hilbert space, combine via the tensor product $\mathcal{H}_{\text{total}} = \mathcal{H}_1 \otimes \mathcal{H}_2$ to form larger, composite dynamical wholes. This mechanism is essential for enabling quintessentially quantum phenomena like **entanglement**, where the state of the whole cannot be factored into independent states of its parts.

The Historical Debate on Completeness

- **The Algorithm’s Claim to Completeness:** The four-postulate quantum algorithm is, in principle, a complete and self-contained logical structure. For over a century, it has provided an astonishingly accurate description of the physical world, from subatomic particles to the chemistry of life.

- **The Core of the Debate: Interpretation of the Quantum State:** The profound question of whether this mathematical description is *complete* has fueled one of the greatest debates in scientific history. This debate crystallized around the physical meaning of the wave function ψ.

- **The Ensemble/Statistical View (Incompleteness):** Physicists like **Einstein, Schrödinger, and de Broglie** argued the algorithm is *incomplete*. They viewed it as a statistical tool for *ensembles* of systems, not a description of individual reality. Einstein famously held that "the wave function cannot be a complete description of reality," believing a deeper, deterministic theory with "hidden variables" must exist.
- **The Copenhagen View (Completeness):** Proponents like **Bohr, Pauli, and von Neumann** maintained the description *is* complete. For them, the wave function represents the full reality of an individual system, and its inherent probabilistic nature is fundamental, not a sign of ignorance.

- **Experimental Verdict (Bell's Inequalities):** Modern experimental evidence, particularly from rigorous tests of **Bell's inequalities**, has strongly vindicated the standard quantum framework. These tests have ruled out broad classes of local hidden variable theories, confirming the non-local correlations predicted by quantum mechanics and thus supporting its completeness at the operational level.

- **Future Directions: Seeking a Deeper Foundation:** Despite its empirical success, the quest for a deeper understanding persists. Many researchers, especially in quantum gravity and foundations, believe quantum mechanics itself may be an **effective theory**—a highly successful approximation emerging from a more fundamental framework (e.g., involving quantum spacetime) yet to be discovered. Therefore, while the quantum algorithm is complete as a *description*, the inquiry into what it implies about the ultimate nature of reality remains a central, driving force in modern physics.

The Algorithmic Completeness

These four postulates provide a complete and self-contained **algorithmic framework** for quantum mechanics. In principle, from these postulates alone, one can derive all standard quantum theory. This framework's power is evident in what it does *not* postulate:

- **The Schrödinger equation**: Emerges from Postulate IIb (unitary evolution) for specific Hamiltonians.
- **The uncertainty principle**: Derivable from Postulates I-III (non-commuting operators).
- **Wave-particle duality**: Contained within the state representation and the measurement postulate.
- **Classical correspondence**: Emerges in the appropriate limits ($\hbar \to 0$, decoherence).

The remainder of this course will be devoted to unpacking these postulates and extracting their physical consequences.

An Important Distinction: It is crucial to distinguish between the *algorithmic completeness* of the framework—its logical self-sufficiency—and the *interpretive completeness* of its central object, the quantum state. Whether the wave function describes an individual system's full reality or merely statistical knowledge of an ensemble has been a central philosophical debate since the theory's inception (Einstein vs. Bohr). While this course will operate within the standard, empirically validated framework, we will remain mindful of this deeper foundational question.

4 Foundations of Quantum Mechanics in the Position Space Representation

We begin our formulation of quantum mechanics for a single particle moving in three-dimensional space, $\mathbb{R}^3$. The state of the particle is completely described by a wavefunction, and its evolution and the results of measurements are governed by a set of fundamental postulates.

4.1 Postulate I: The Quantum State

The first postulate establishes the mathematical object that contains all information about a physical system.

Postulate I: The Quantum State

The quantum state of a single particle is completely described by a complex-valued function $\psi(\mathbf{x}, t)$, known as the **wavefunction**. At a fixed time, say $t = 0$, the wavefunction $\psi(\mathbf{x}) \equiv \psi(\mathbf{x}, 0)$ is an element of a complex Hilbert space $\mathcal{H}$.

For a particle in space, the relevant Hilbert space is the space of square-integrable functions, denoted $L^2(\mathbb{R}^3; \mathbb{C})$.

$$\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C}) = \left\{ \psi : \mathbb{R}^3 \to \mathbb{C} \;\middle|\; \int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x < \infty \right\}. \tag{4}$$

The inner product on this space, which maps two wavefunctions to a complex number, is defined by the integral

$$\langle \phi | \psi \rangle = \int_{\mathbb{R}^3} \phi^*(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3x. \tag{5}$$

The norm of a state is given by

$$||\psi|| = \sqrt{\langle \psi | \psi \rangle}. \tag{6}$$

This norm must be finite. States are typically normalized such that $||\psi|| = 1$, a condition that will be linked to probability by the Born rule.

Remark 4.1 (The Superposition Principle)**.** If $\psi_1(\mathbf{x})$ and $\psi_2(\mathbf{x})$ are possible quantum states of a system, then any linear combination

$$\psi(\mathbf{x}) = c_1\psi_1(\mathbf{x}) + c_2\psi_2(\mathbf{x}) \tag{7}$$

where $c_1, c_2 \in \mathbb{C}$, is also a possible quantum state. The physical justification for this principle will become evident when we discuss the linearity of the Schrödinger equation.

4.2 Postulate II: Operators, Observables, and Time Evolution

This postulate introduces the mathematical objects that represent physical quantities and the law that governs the change of the quantum state over time.

4.2.1 II.a Physical Observables as Hermitian Operators

Physical observables are represented by linear operators acting on the Hilbert space $L^2(\mathbb{R}^3; \mathbb{C})$.

Definition 12 (Linear Operator)**.** A linear operator $\hat{A}$ is a map from $\mathcal{H}$ to $\mathcal{H}$ such that for any $\psi, \phi \in \mathcal{H}$ and $c \in \mathbb{C}$,

$$\hat{A}(c\psi + \phi) = c\hat{A}\psi + \hat{A}\phi. \tag{8}$$

Here, ψ and ϕ are functions of the position variable $\mathbf{x}$, i.e., $\psi \equiv \psi(\mathbf{x})$.

Definition 13 (Adjoint Operator)**.** The adjoint of an operator $\hat{A}$, denoted $\hat{A}^\dagger$, is defined by the relation

$$\langle\phi|\hat{A}\psi\rangle = \langle\hat{A}^\dagger\phi|\psi\rangle, \tag{9}$$

which must hold for all $\psi, \phi \in \mathcal{H}$. In the integral form, this is

$$\int_{\mathbb{R}^3} \phi^*(\mathbf{x})(\hat{A}\psi)(\mathbf{x})\mathrm{d}^3x = \int_{\mathbb{R}^3} (\hat{A}^\dagger\phi)^*(\mathbf{x})\psi(\mathbf{x})\mathrm{d}^3x. \tag{10}$$

The adjoint operation has the following important properties (for operators $\hat{A}$, $\hat{B}$ and a complex constant c):

- $(\hat{A}^\dagger)^\dagger = \hat{A}$
- $(c\hat{A})^\dagger = c^*\hat{A}^\dagger$
- $(\hat{A}+\hat{B})^\dagger = \hat{A}^\dagger + \hat{B}^\dagger$
- $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger\hat{A}^\dagger$

Definition 14 (Hermitian Operator)**.** An operator $\hat{A}$ is **Hermitian** (or self-adjoint) if it is equal to its own adjoint: $\hat{A}^\dagger = \hat{A}$. This implies that for all $\psi, \phi \in \mathcal{H}$,

$$\langle\phi|\hat{A}\psi\rangle = \langle\hat{A}\phi|\psi\rangle. \tag{11}$$

The fundamental observables of a single particle are represented by the following Hermitian operators.

Definition 15 (Position Operator)**.** In three dimensions, the position operator $\hat{\mathbf{x}} = (\hat{x}, \hat{y}, \hat{z})$ acts on a wavefunction by multiplication with the coordinate:

$$(\hat{x}\psi)(\mathbf{x}) = x\psi(\mathbf{x}), \tag{12}$$

$$(\hat{y}\psi)(\mathbf{x}) = y\psi(\mathbf{x}), \tag{13}$$

$$(\hat{z}\psi)(\mathbf{x}) = z\psi(\mathbf{x}). \tag{14}$$

In one dimension, this reduces to $(\hat{x}\psi)(x) = x\psi(x)$.

Definition 16 (Momentum Operator)**.** In three dimensions, the momentum operator $\hat{\mathbf{p}} = (\hat{p}_x, \hat{p}_y, \hat{p}_z)$ is defined as a differential operator:

$$(\hat{p}_x\psi)(\mathbf{x}) = -i\hbar\frac{\partial}{\partial x}\psi(\mathbf{x}), \tag{15}$$

$$(\hat{p}_y\psi)(\mathbf{x}) = -i\hbar\frac{\partial}{\partial y}\psi(\mathbf{x}), \tag{16}$$

$$(\hat{p}_z\psi)(\mathbf{x}) = -i\hbar\frac{\partial}{\partial z}\psi(\mathbf{x}). \tag{17}$$

In one dimension, this reduces to

$$(\hat{p}\psi)(x) = -i\hbar\frac{\mathrm{d}}{\mathrm{d}x}\psi(x). \tag{18}$$

The Hermiticity of these operators can be shown using the inner product. For the position operator in 1D:

$$\langle\phi|\hat{x}\psi\rangle = \int_{-\infty}^{\infty} \phi^*(x)\,x\,\psi(x)\mathrm{d}x = \int_{-\infty}^{\infty} (x\phi(x))^*\psi(x)\mathrm{d}x = \langle\hat{x}\phi|\psi\rangle, \tag{19}$$

confirming $\hat{x}^\dagger = \hat{x}$. For the momentum operator in 1D, we require integration by parts:

$$\langle \phi | \hat{p}\psi \rangle = \int_{-\infty}^{\infty} \phi^*(x) \left(-i\hbar \frac{\mathrm{d}\psi}{\mathrm{d}x} \right) \mathrm{d}x \tag{20}$$

$$= [-i\hbar \phi^*(x)\psi(x)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \left(-i\hbar \frac{\mathrm{d}\phi}{\mathrm{d}x} \right)^* \psi(x)\mathrm{d}x. \tag{21}$$

For the operator to be Hermitian, the boundary term must vanish. This imposes a condition on the wavefunctions in our Hilbert space:

$$\lim_{x \to \pm\infty} \psi(x) = 0, \tag{22}$$

which is satisfied for square-integrable functions. With this, we get $\langle \phi | \hat{p}\psi \rangle = \langle \hat{p}\phi | \psi \rangle$, so $\hat{p}^\dagger = \hat{p}$.

4.2.2 II.b Time Evolution: The Schrödinger Equation

The time evolution of a quantum state is governed by the Schrödinger equation.

Postulate II.b: The Schrödinger Equation

The time evolution of the wavefunction $\psi(\mathbf{x}, t)$ is determined by the **time-dependent Schrödinger equation**:

$$\boxed{i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \hat{H} \psi(\mathbf{x}, t)}, \tag{23}$$

where $\hat{H}$ is the Hamiltonian operator representing the total energy of the system.

Illustration 1: The Time-Evolution Operator. For a system with a time-independent Hamiltonian $\hat{H}$, we can solve the Schrödinger equation using the method of separation of variables. This is a standard technique for solving partial differential equations, where we assume the solution can be written as a product of a function of space alone and a function of time alone:

$$\psi(\mathbf{x}, t) = \psi(\mathbf{x})T(t). \tag{24}$$

Substituting this into the time-dependent Schrödinger equation gives:

$$i\hbar \psi(\mathbf{x}) \frac{\mathrm{d}T}{\mathrm{d}t} = T(t)\hat{H}\psi(\mathbf{x}). \tag{25}$$

Now, divide both sides by $\psi(\mathbf{x})T(t)$ (where the functions are non-zero):

$$i\hbar \frac{1}{T(t)} \frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{\psi(\mathbf{x})} \hat{H}\psi(\mathbf{x}). \tag{26}$$

The left-hand side is a function of time only, while the right-hand side is a function of position only. For these to be equal for all $\mathbf{x}$ and t, they must both be equal to a constant, which we call E (which will turn out to be the energy). This yields two separate equations:

$$i\hbar \frac{\mathrm{d}T}{\mathrm{d}t} = ET(t), \tag{27}$$

$$\hat{H}\psi(\mathbf{x}) = E\psi(\mathbf{x}). \tag{28}$$

Equation 27 is an ordinary differential equation for time with the solution

$$T(t) = e^{-iEt/\hbar}. \tag{29}$$

Equation 28 is the time-independent Schrödinger equation, an eigenvalue equation for the Hamiltonian. The general solution to the original equation can then be written using the **time-evolution operator** $\hat{U}(t, t_0 = 0)$:

$$\psi(\mathbf{x}, t) = e^{-i\hat{H}t/\hbar}\psi(\mathbf{x}, 0) \equiv \hat{U}(t, 0)\psi(\mathbf{x}, 0). \tag{30}$$

The operator $\hat{U}$ is unitary, meaning $\hat{U}^\dagger = \hat{U}^{-1}$, so it preserves the norm of the state. It also satisfies the group properties

$$\hat{U}(t, t) = 1, \tag{31}$$

and

$$\hat{U}(t_2, t_1)\hat{U}(t_1, t_0) = \hat{U}(t_2, t_0). \tag{32}$$

Illustration 2: Time-Independent Schrödinger Equation. As derived above, when the Hamiltonian is time-independent, separating the variables leads directly to the time-independent Schrödinger equation:

$$\boxed{\hat{H}\phi(\mathbf{x}) = E\phi(\mathbf{x})}. \tag{33}$$

Here, E are the allowed energy eigenvalues and $\phi(\mathbf{x})$ are the corresponding energy eigenstates. The full time-dependent solution for an energy eigenstate is then $\phi(\mathbf{x})e^{-iEt/\hbar}$.

4.3 Postulate III: The Measurement Postulate and the Born Rule

Before stating the measurement postulate, we must first understand a fundamental mathematical fact about Hermitian operators and their eigenfunctions.

4.3.1 Mathematical Prelude: Eigenfunctions as Basis Vectors

Recall from linear algebra that in a finite-dimensional vector space, a Hermitian matrix has a complete set of orthonormal eigenvectors that form a basis. Every vector can be uniquely expanded as a linear combination of these eigenvectors. The spectral theorem extends this crucial property to Hermitian operators on the infinite-dimensional Hilbert space $L^2(\mathbb{R}^3; \mathbb{C})$.

Theorem 3 (Spectral Theorem)**.** Let $\hat{A}$ be a Hermitian operator on $\mathcal{H} = L^2(\mathbb{R}^3; \mathbb{C})$ representing a physical observable. Then the eigenfunctions $\{\varphi_n(\mathbf{x})\}$ of $\hat{A}$, corresponding to eigenvalues a_n, form a **complete orthonormal basis** for the Hilbert space.

This theorem has two essential consequences, expressed mathematically in the language of wavefunctions:

1. **Orthonormality**: Different eigenfunctions are orthogonal, and each eigenfunction is normalized. For a discrete spectrum (like bound states in an atom), this is written as:

$$\langle \varphi_m | \varphi_n \rangle = \int_{\mathbb{R}^3} \varphi_m^*(\mathbf{x})\, \varphi_n(\mathbf{x})\, \mathrm{d}^3x = \delta_{mn}, \tag{34}$$

where δ_{mn} is the Kronecker delta ($= 1$ if $m = n$, 0 otherwise).

For operators with a continuous spectrum (such as the momentum operator, which we will study in Application 3), the orthonormality condition involves the Dirac delta function:

$$\int_{\mathbb{R}^3} \varphi_{p'}^*(\mathbf{x})\, \varphi_p(\mathbf{x})\, \mathrm{d}^3x = \delta(p' - p). \tag{35}$$

We will explore the physical meaning of this when we encounter continuous spectra later.

2. **Completeness**: Any wavefunction $\psi(\mathbf{x})$ in the Hilbert space can be expanded as a linear combination of the eigenfunctions. For a discrete spectrum:

$$\psi(\mathbf{x}) = \sum_n c_n \, \varphi_n(\mathbf{x}), \tag{36}$$

where the expansion coefficients c_n are complex numbers given by the inner product:

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x}) \, \psi(\mathbf{x}) \, \mathrm{d}^3 x. \tag{37}$$

For a continuous spectrum, the sum becomes an integral:

$$\psi(\mathbf{x}) = \int c(p) \, \varphi_p(\mathbf{x}) \, \mathrm{d}p, \tag{38}$$

with $c(p) = \langle \varphi_p | \psi \rangle$.

The completeness relation can also be expressed as a condition on the eigenfunctions themselves. In the discrete case:

$$\sum_n \varphi_n^*(\mathbf{x}') \, \varphi_n(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{39}$$

which means that any wavefunction can be reconstructed by integrating over its expansion:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \left(\sum_n \varphi_n^*(\mathbf{x}') \varphi_n(\mathbf{x}) \right) \psi(\mathbf{x}') \, \mathrm{d}^3 x'. \tag{40}$$

Remark 4.2 (Looking Ahead)**.** In Applications 3 and 4, we will encounter two important examples of such eigenfunction bases:

- The **momentum eigenfunctions** $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$, which form a continuous basis and lead to the momentum space representation via Fourier transforms.
- The **position eigenfunctions** $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$, which are generalized functions (distributions) that form a continuous basis and directly yield the Born rule probability density $|\psi(\mathbf{x})|^2$.

Both are special cases of the general spectral theorem stated above, and their properties will be derived in detail later.

4.3.2 The Measurement Postulate

With this mathematical framework in place, we can now state the measurement postulate. We assume the wavefunction is normalized:

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \, \mathrm{d}^3 x = 1. \tag{41}$$

Consider an observable represented by a Hermitian operator $\hat{A}$. Let $\{\varphi_n(\mathbf{x})\}$ be the complete set of eigenfunctions of $\hat{A}$ with corresponding eigenvalues a_n, satisfying:

$$\hat{A}\varphi_n = a_n \varphi_n. \tag{42}$$

Since the eigenfunctions form a complete orthonormal basis, any state $\psi(\mathbf{x})$ can be expanded as:

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{43}$$

where the expansion coefficients are given by:

$$c_n = \langle \varphi_n | \psi \rangle = \int_{\mathbb{R}^3} \varphi_n^*(\mathbf{x}) \, \psi(\mathbf{x}) \, \mathrm{d}^3 x. \tag{44}$$

Postulate III: Measurement

If a quantum system is in a state described by the normalized wavefunction

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{45}$$

then a measurement of the observable corresponding to $\hat{A}$ will yield:

1. **Outcome**: One of the eigenvalues a_n of $\hat{A}$.
2. **Probability**: The probability of obtaining a specific eigenvalue a_n is given by the square of the magnitude of its expansion coefficient,
$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{46}$$
3. **State Collapse**: Immediately after the measurement, the quantum state of the system collapses from $\psi(\mathbf{x})$ to the corresponding eigenfunction $\varphi_n(\mathbf{x})$.

If the measured eigenfunction is not normalized, the post-measurement state must be renormalized.

Remark 4.3. It is crucial to distinguish the three fundamental quantities involved:

- **Quantum State**: $\varphi_n(\mathbf{x})$ (a function in L^2).
- **Measurement Outcome**: a_n (a real number).
- **Probability Amplitude**: $c_n = \langle \varphi_n | \psi \rangle$ (a complex number).
- **Probability**: $\mathbb{P}(A = a_n) = |c_n|^2$ (a real number between 0 and 1).

Remark 4.4 (Continuous Spectra)**.** For operators with a continuous spectrum (such as position or momentum), the expansion becomes an integral and the probability becomes a probability density. For example, the position operator $\hat{\mathbf{x}}$ has eigenfunctions $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$, and the expansion coefficient for a state $\psi(\mathbf{x})$ is simply $c(\mathbf{x}') = \psi(\mathbf{x}')$. This leads naturally to the Born rule for position measurements, which we state next.

Postulate III: Measurement

If a quantum system is in a state described by the normalized wavefunction

$$\psi(\mathbf{x}) = \sum_n c_n \varphi_n(\mathbf{x}), \tag{47}$$

then a measurement of the observable corresponding to $\hat{A}$ will yield:

1. **Outcome**: One of the eigenvalues a_n of $\hat{A}$.
2. **Probability**: The probability of obtaining a specific eigenvalue a_n is given by the square of the magnitude of its expansion coefficient,
$$\mathbb{P}(A = a_n) = |c_n|^2. \tag{48}$$
3. **State Collapse**: Immediately after the measurement, the quantum state of the system collapses from $\psi(\mathbf{x})$ to the corresponding eigenfunction $\varphi_n(\mathbf{x})$.

If the measured eigenfunction is not normalized, the post-measurement state must be renormalized.

Remark 4.5. It is crucial to distinguish the three fundamental quantities involved:

- **Quantum State**: $\varphi_n(\mathbf{x})$ (a function in L^2).
- **Measurement Outcome**: a_n (a real number).
- **Probability Amplitude**: $c_n = \langle \varphi_n | \psi \rangle$ (a complex number).
- **Probability**: $\mathbb{P}(A = a_n) = |c_n|^2$ (a real number between 0 and 1).

4.3.3 Interlude: A Brief Review of Probability Concepts

To better understand the measurement postulate, let us review some basic ideas from probability theory. Consider a discrete random variable X that can take on values $x_1, x_2, \ldots, x_n$ with probabilities $p_1, p_2, \ldots, p_n$. These probabilities satisfy

$$\sum_{i=1}^{n} p_i = 1. \tag{49}$$

Definition 17 (Average Value (Mean))**.** The average value (or mean) of X, denoted $\langle X \rangle$, is calculated by summing each possible value multiplied by its probability:

$$\langle X \rangle = \sum_{i=1}^{n} x_i p_i. \tag{50}$$

This represents the value we expect to get on average if we perform the measurement many times.

Definition 18 (Variance and Standard Deviation)**.** The variance, σ_X^2, measures the spread of the possible outcomes around the mean. It is defined as the average of the squared deviation from the mean:

$$\sigma_X^2 = \langle (X - \langle X \rangle)^2 \rangle = \sum_{i=1}^{n} (x_i - \langle X \rangle)^2 p_i. \tag{51}$$

The variance can also be expressed in a form that is often easier to compute:

$$\sigma_X^2 = \langle X^2 - 2X\langle X \rangle + \langle X \rangle^2 \rangle \tag{52}$$

$$= \langle X^2 \rangle - 2\langle X \rangle \langle X \rangle + \langle X \rangle^2 \tag{53}$$

$$= \langle X^2 \rangle - \langle X \rangle^2. \tag{54}$$

The **standard deviation** σ_X (often denoted ΔX) is the square root of the variance:

$$\Delta X = \sigma_X = \sqrt{\langle X^2 \rangle - \langle X \rangle^2}. \tag{55}$$

The standard deviation gives a measure of the uncertainty in the outcome of a measurement.

We can now apply these concepts directly to quantum mechanics. When we measure an observable $\hat{A}$ on a state ψ, the eigenvalues a_n are the possible outcomes, and the probabilities are $|c_n|^2$.

Definition 19 (Expectation Value in Quantum Mechanics)**.** The expected value (or mean value) of many measurements of the observable $\hat{A}$ on a system in the state $\psi(\mathbf{x})$ is given by

$$\langle \hat{A} \rangle_\psi = \sum_n a_n |c_n|^2. \tag{56}$$

This can be expressed more directly in terms of the wavefunction using the inner product:

$$\boxed{\langle \hat{A} \rangle_\psi = \langle \psi | \hat{A} | \psi \rangle = \int_{\mathbb{R}^3} \psi^*(\mathbf{x}) (\hat{A}\psi)(\mathbf{x}) \mathrm{d}^3 x}. \tag{57}$$

Definition 20 (Standard Deviation of an Operator)**.** For a system in a state ψ, the standard deviation (or uncertainty) ΔA of an observable represented by the Hermitian operator $\hat{A}$ is defined, in analogy with the classical definition, as

$$\Delta A = \sqrt{\langle (\hat{A} - \langle \hat{A} \rangle)^2 \rangle}. \tag{58}$$

This can be expanded to the equivalent and more practical form:

$$\boxed{\Delta A = \sqrt{\langle \hat{A}^2 \rangle_\psi - \langle \hat{A} \rangle_\psi^2}}. \tag{59}$$

This quantity represents the root-mean-square deviation of measurement outcomes from the average value.

Definition 21 (Born Rule)**.** The probability density for finding a particle in a state $\psi(\mathbf{x})$ at position $\mathbf{x}$ is $|\psi(\mathbf{x})|^2$. More formally, the probability of finding the particle in a region $D \subset \mathbb{R}^3$ is

$$\boxed{\mathbb{P}(\mathbf{x} \in D) = \int_D |\psi(\mathbf{x})|^2 \mathrm{d}^3 x}. \tag{60}$$

4.4 Applications and Further Topics

4.4.1 Application 1: Commutators and the Uncertainty Principle

The order in which operators act on a state is significant. This is captured by the commutator.

Definition 22 (Commutator)**.** The commutator of two operators $\hat{A}$ and $\hat{B}$ is defined as

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}. \tag{61}$$

The commutator is linear and satisfies important identities such as

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]. \tag{62}$$

Definition 23 (Canonical Commutation Relations (CCR))**.** The fundamental commutation relation between position and momentum in three dimensions is:

$$\boxed{[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}}, \tag{63}$$

where $i, j = x, y, z$. In one dimension, this is

$$[\hat{x}, \hat{p}] = i\hbar. \tag{64}$$

We are now in a position to state the most famous uncertainty principle.

Theorem 4 (Generalized Uncertainty Principle)**.** For any two Hermitian operators $\hat{A}$ and $\hat{B}$ representing physical observables, the product of their standard deviations in any quantum state ψ obeys the Robertson-Schrödinger uncertainty relation:

$$\boxed{(\Delta A)^2(\Delta B)^2 \geq \left| \frac{1}{2i} \langle [\hat{A}, \hat{B}] \rangle \right|^2}. \tag{65}$$

A simpler and more common form, which follows from the above, is:

$$\boxed{\Delta A \, \Delta B \geq \frac{1}{2} \left| \langle [\hat{A}, \hat{B}] \rangle \right|}. \tag{66}$$

This inequality shows that if two operators do not commute ($[\hat{A}, \hat{B}] \neq 0$), there is a fundamental lower bound on the product of their uncertainties. It is impossible to prepare a state in which both observables have arbitrarily precise values simultaneously.

We now apply this general principle to the specific case of position and momentum.

Corollary 4.1 (Heisenberg Uncertainty Principle for Position and Momentum)**.** For a single particle in three dimensions, the canonical commutation relations are:

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}, \tag{67}$$

where $i, j = x, y, z$. Applying the generalized uncertainty principle yields the famous Heisenberg uncertainty relations:

$$\Delta x\, \Delta p_x \geq \frac{\hbar}{2}, \tag{68}$$

$$\Delta y\, \Delta p_y \geq \frac{\hbar}{2}, \tag{69}$$

$$\Delta z\, \Delta p_z \geq \frac{\hbar}{2}. \tag{70}$$

It is crucial to note that for different components, e.g., $\hat{x}$ and $\hat{p}_y$, the commutator vanishes:

$$[\hat{x}, \hat{p}_y] = 0. \tag{71}$$

Consequently, there is no fundamental quantum limit preventing the simultaneous precise knowledge of, say, the x-position and the y-momentum of a particle. Their uncertainties can both be zero in principle.

In one dimension, the relations reduce to the single, most commonly quoted form:

$$\boxed{\Delta x\, \Delta p \geq \frac{\hbar}{2}}. \tag{72}$$

Remark 4.6 (Physical Meaning)**.** The Heisenberg uncertainty principle is not a statement about the limitations of our measurement devices, but a fundamental property of quantum states. A particle simply cannot possess both a well-defined position and a well-defined momentum. This is a direct consequence of the wave-like nature of matter: a wave confined to a small region in space (small Δx) is composed of a wide range of wavelengths (large Δp), and vice versa.

Physically, this principle implies that it is impossible to simultaneously know the exact position and exact momentum of a particle. A state cannot be a simultaneous eigenstate of both $\hat{x}$ and $\hat{p}$ because their operators do not commute.

An important example of commuting operators is $[\hat{p}, \hat{H}] = 0$ for a free particle. This indicates that momentum and energy can be measured simultaneously in such a system.

4.4.2 Application 2: Dynamics of a Free Particle

The Hamiltonian for a single particle of mass m in a potential $V(\mathbf{x})$ is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\mathbf{x}) = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}). \tag{73}$$

For a **free particle**, $V(\mathbf{x}) = 0$, and the time-dependent Schrödinger equation becomes:

$$i\hbar\frac{\partial}{\partial t}\psi(\mathbf{x}, t) = -\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{x}, t). \tag{74}$$

We seek to derive the fundamental wave relations from this equation and our operator definitions.

1. **De Broglie Relation:** First, consider the eigenvalue problem for the momentum operator, $\hat{\mathbf{p}}\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x})$. Using $\hat{\mathbf{p}} = -i\hbar\nabla$, we have:

$$-i\hbar\nabla\varphi_{\mathbf{p}}(\mathbf{x}) = \mathbf{p}\varphi_{\mathbf{p}}(\mathbf{x}). \tag{75}$$

This is a differential equation whose solution is an exponential function. In three dimensions, the solution is:

$$\varphi_{\mathbf{p}}(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}}e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}. \tag{76}$$

By comparing this to a standard plane wave of the form $e^{i\mathbf{k}\cdot\mathbf{x}}$, we identify the wave vector $\mathbf{k}$ as $\mathbf{k} = \mathbf{p}/\hbar$. This yields the de Broglie relation:

$$\boxed{\mathbf{p} = \hbar\mathbf{k}}. \tag{77}$$

2. **Planck Relation and Dispersion:** Now, we look for a solution to the time-dependent equation (74) that is consistent with this. We try a plane wave solution that includes the time dependence we found from separation of variables:

$$\psi(\mathbf{x}, t) = Ae^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)}. \tag{78}$$

Substituting this into the left-hand side of (74) gives:

$$i\hbar\frac{\partial}{\partial t}\psi = i\hbar(-i\omega)\psi = \hbar\omega\psi. \tag{79}$$

Substituting into the right-hand side gives:

$$-\frac{\hbar^2}{2m}\nabla^2\psi = -\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}\right)e^{i(\mathbf{k}\cdot\mathbf{x}-\omega t)} \tag{80}$$

$$= -\frac{\hbar^2}{2m}(i^2k_x^2 + i^2k_y^2 + i^2k_z^2)\psi \tag{81}$$

$$= -\frac{\hbar^2}{2m}(-k_x^2 - k_y^2 - k_z^2)\psi \tag{82}$$

$$= \frac{\hbar^2k^2}{2m}\psi. \tag{83}$$

For the equation to hold, the coefficients of ψ on both sides must be equal:

$$\hbar\omega = \frac{\hbar^2k^2}{2m}. \tag{84}$$

Recognizing the right-hand side as the kinetic energy $E = p^2/2m = \hbar^2k^2/2m$, we obtain the Planck relation and the dispersion relation:

$$\boxed{E = \hbar\omega} \quad \text{and} \quad \boxed{E = \frac{\hbar^2k^2}{2m}}. \tag{85}$$

We see that the definitions of the position and momentum operators, combined with the Schrödinger equation, directly yield the fundamental relations of wave-particle duality.

4.4.3 Application 3: Momentum Space Representation

The momentum eigenfunctions $\varphi_{\mathbf{p}}(\mathbf{x})$ form a complete basis. This means that any state $\psi(\mathbf{x})$ can be expressed as a linear combination of these basis functions. Because the momentum eigenvalues $\mathbf{p}$ are continuous, this expansion takes the form of an integral rather than a sum.

We begin by writing the expansion as

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \varphi_{\mathbf{p}}(\mathbf{x})\, \mathrm{d}^3 p. \tag{86}$$

Here, $c(\mathbf{p})$ is a complex coefficient that tells us how much of the momentum eigenstate $\mathbf{p}$ is contained in $\psi(\mathbf{x})$. Substituting the explicit form of the momentum eigenfunction, we get:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} c(\mathbf{p})\, \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p. \tag{87}$$

It is convenient to define a new function $\tilde{\psi}(\mathbf{p})$, called the **momentum space wavefunction**, which is essentially the density of the coefficient $c(\mathbf{p})$. We set

$$\tilde{\psi}(\mathbf{p}) = (2\pi\hbar)^{3/2}\, c(\mathbf{p}). \tag{88}$$

Substituting this definition into equation (87), the factors of $(2\pi\hbar)^{3/2}$ cancel, yielding:

$$\boxed{\psi(\mathbf{x}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p})\, e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 p}. \tag{89}$$

This expression is precisely the **inverse Fourier transform** of $\tilde{\psi}(\mathbf{p})$.

To find the function $\tilde{\psi}(\mathbf{p})$ from $\psi(\mathbf{x})$, we need the inverse relation. This is obtained from the orthonormality of the momentum eigenfunctions:

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \varphi_{\mathbf{p}}(\mathbf{x}) \mathrm{d}^3 x = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{90}$$

Now, take the inner product of both sides of the expansion in equation (87) with a specific eigenfunction $\varphi_{\mathbf{p}'}(\mathbf{x})$:

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \int_{\mathbb{R}^3} \varphi^*_{\mathbf{p}'}(\mathbf{x}) \psi(\mathbf{x}) \mathrm{d}^3 x & (91)\\ &= \int_{\mathbb{R}^3} \left(\frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar} \right) \left(\frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3 p \right) \mathrm{d}^3 x. & (92) \end{aligned}$$

Interchanging the order of integration and using the integral representation of the delta function,

$$\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x = (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}'), \tag{93}$$

we obtain:

$$\begin{aligned} \langle \varphi_{\mathbf{p}'} | \psi \rangle &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) \left(\int_{\mathbb{R}^3} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \mathrm{d}^3 x \right) \mathrm{d}^3 p & (94)\\ &= \frac{1}{(2\pi\hbar)^3} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) (2\pi\hbar)^3 \delta^{(3)}(\mathbf{p} - \mathbf{p}') \mathrm{d}^3 p & (95)\\ &= \tilde{\psi}(\mathbf{p}'). & (96) \end{aligned}$$

Since the left-hand side is just the inner product, we arrive at the **forward Fourier transform** relation:

$$\boxed{\tilde{\psi}(\mathbf{p}) = \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \psi(\mathbf{x})\, e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar}\, \mathrm{d}^3 x}. \tag{97}$$

A key property of the Fourier transform is that it preserves the norm of the function (Plancherel's theorem):

$$\int_{\mathbb{R}^3} |\psi(\mathbf{x})|^2 \mathrm{d}^3x = \int_{\mathbb{R}^3} \left| \frac{1}{(2\pi\hbar)^{3/2}} \int_{\mathbb{R}^3} \tilde{\psi}(\mathbf{p}) e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \mathrm{d}^3p \right|^2 \mathrm{d}^3x \tag{98}$$

$$= \int_{\mathbb{R}^3} |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p. \tag{99}$$

This confirms that if $\psi(\mathbf{x})$ is normalized, then $\tilde{\psi}(\mathbf{p})$ is also normalized. Therefore, $|\tilde{\psi}(\mathbf{p})|^2$ can be interpreted as a probability density in momentum space. The probability of measuring the particle's momentum in a region D of momentum space is:

$$\boxed{\mathbb{P}(\mathbf{p} \in D) = \int_D |\tilde{\psi}(\mathbf{p})|^2 \mathrm{d}^3p}. \tag{100}$$

Remark 4.7 (Normalization of Plane Waves)**.** The momentum eigenfunction $\varphi_{\mathbf{p}}(\mathbf{x}) = (2\pi\hbar)^{-3/2} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ is not square-integrable, meaning it is not a member of L^2. Its normalization is to a Dirac delta function, as we used above:

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{101}$$

4.4.4 Application 4: Position Eigenstates

We now seek the eigenfunctions of the position operator $\hat{\mathbf{x}}$. Let $\varphi_{\mathbf{x}'}(\mathbf{x})$ be an eigenstate with eigenvalue $\mathbf{x}'$:

$$\hat{\mathbf{x}} \varphi_{\mathbf{x}'}(\mathbf{x}) = \mathbf{x}' \varphi_{\mathbf{x}'}(\mathbf{x}). \tag{102}$$

Recall the sifting property of the Dirac delta function, which allows any function to be expanded as:

$$\psi(\mathbf{x}) = \int_{\mathbb{R}^3} \psi(\mathbf{x}') \delta^{(3)}(\mathbf{x} - \mathbf{x}') \mathrm{d}^3x'. \tag{103}$$

This expansion suggests that the delta functions $\delta^{(3)}(\mathbf{x} - \mathbf{x}')$ act as a basis for expanding wavefunctions. To verify they are eigenfunctions, we substitute $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ into the eigenvalue equation:

$$\hat{\mathbf{x}} \delta^{(3)}(\mathbf{x} - \mathbf{x}') = \mathbf{x} \delta^{(3)}(\mathbf{x} - \mathbf{x}') \tag{104}$$

$$= \mathbf{x}' \delta^{(3)}(\mathbf{x} - \mathbf{x}'), \tag{105}$$

where the last equality uses the defining property of the delta function: $f(\mathbf{x})\delta(\mathbf{x} - \mathbf{x}') = f(\mathbf{x}')\delta(\mathbf{x} - \mathbf{x}')$. Thus, the (generalized) eigenfunctions of the position operator are

$$\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}'). \tag{106}$$

Remark 4.8 (Normalization of Delta Functions)**.** These position eigenfunctions are also not square-integrable. They are normalized to a delta function in their eigenvalues:

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \int_{\mathbb{R}^3} \delta^{(3)}(\mathbf{x} - \mathbf{x}')\, \delta^{(3)}(\mathbf{x} - \mathbf{x}'')\, \mathrm{d}^3x = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{107}$$

This means that while the position eigenfunctions are not normalizable in the usual sense (their norm is infinite), they satisfy a weaker form of orthonormality: the inner product of two such eigenfunctions is zero unless their eigenvalues are identical, in which case it is infinite in a controlled way represented by the Dirac delta function.

Remark 4.9 (Comparison with Momentum Eigenfunctions)**.** It is instructive to compare the normalization of position and momentum eigenfunctions:

- **Position eigenfunctions**: $\varphi_{\mathbf{x}'}(\mathbf{x}) = \delta^{(3)}(\mathbf{x} - \mathbf{x}')$ with normalization

$$\langle \varphi_{\mathbf{x}'} | \varphi_{\mathbf{x}''} \rangle = \delta^{(3)}(\mathbf{x}' - \mathbf{x}''). \tag{108}$$

- **Momentum eigenfunctions**: $\varphi_{\mathbf{p}}(\mathbf{x}) = \dfrac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$ with normalization

$$\langle \varphi_{\mathbf{p}'} | \varphi_{\mathbf{p}} \rangle = \delta^{(3)}(\mathbf{p}' - \mathbf{p}). \tag{109}$$

In both cases, the eigenfunctions are not elements of L^2 (they are not square-integrable), but they form a complete basis in the sense of Dirac delta function normalization. This is why they are often called **generalized eigenfunctions**.

Key Differences from Classical Physics

Aspect	Classical Physics	Quantum Mechanics
State Description	Position & momentum (x, p)	Wave function $\Psi(x, t)$
Determinism	Completely deterministic	Probabilistic predictions
Observables	Continuous values	Quantized/discrete values
Measurement	Passive observation	Active disturbance
Superposition	Not allowed	Fundamental principle

Quantum Operators Dictionary

Observable	Classical	Quantum Operator	Eigenvalue Eqn
Position	x	$\hat{x} = x$	$\hat{x}\psi = x_0\psi$
Momentum	p_x	$\hat{p}_x = -i\hbar\frac{\partial}{\partial x}$	$\hat{p}_x\psi = p\psi$
Kinetic Energy	$\frac{p^2}{2m}$	$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2$	$\hat{T}\psi = E\psi$
Potential Energy	$V(x)$	$\hat{V} = V(x)$	$\hat{V}\psi = V(x)\psi$
Total Energy	$H = T + V$	$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x)$	$\hat{H}\psi = E\psi$
Angular Momentum	$\mathbf{L} = \mathbf{r} \times \mathbf{p}$	$\hat{L}_z = -i\hbar\frac{\partial}{\partial \phi}$	$\hat{L}_z\psi = m\hbar\psi$

5 The Historical Approach to Quantum Mechanics

Quantum mechanics emerged from experimental observations that classical physics couldn't explain. The development occurred in two main phases:

Old Quantum Theory (1900-1925)
Planck, Einstein, Bohr, Sommerfeld

$\downarrow$ **Matrix & Wave Mechanics**

Modern Quantum Mechanics (1925-1930)
Heisenberg, Schrödinger, Dirac, Born, Jordan, Pauli

5.1 The Crisis of Classical Physics (Late 19th Century)

By the late 1800s, classical physics (Newtonian mechanics, Maxwell's electromagnetism, thermodynamics) appeared complete. However, several phenomena defied classical explanation:

Unsolved Problems in 1900

1. **Blackbody Radiation**: Spectrum couldn't be explained by classical electrodynamics
2. **Photoelectric Effect**: Light behaving as particles
3. **Atomic Spectra**: Discrete emission/absorption lines
4. **Specific Heat of Solids**: Deviations from Dulong-Petit law at low temperatures
5. **Stability of Atoms**: Why don't electrons spiral into the nucleus?

In what follows, we will discuss some of these historically importan problems in details.

5.1.1 Blackbody Radiation and Planck's Quantum Hypothesis (1900)

Definition 24 (Blackbody)**.** *An ideal object that absorbs all radiation incident upon it and emits radiation with a characteristic spectrum that depends only on its temperature.*

The classical Rayleigh-Jeans law predicted the **ultraviolet catastrophe**:

$$\rho_{\text{classical}}(\nu, T) = \frac{8\pi\nu^2}{c^3} k_B T \quad \xrightarrow{\nu \to \infty} \quad \infty!$$

Planck's Radical Assumption (December 1900)

Energy of electromagnetic oscillators is **quantized**:

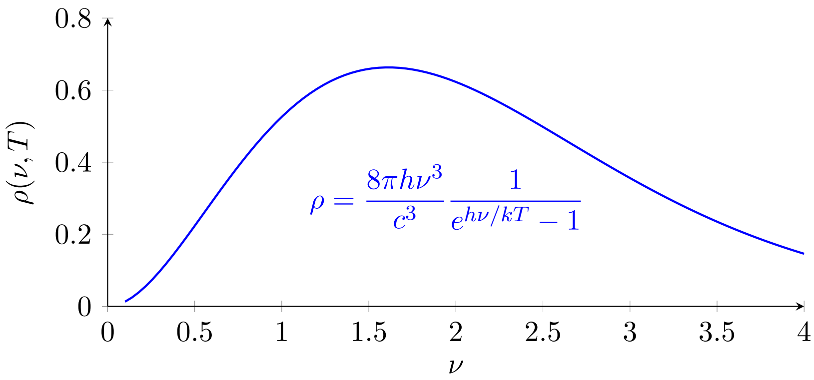
$$E = nh\nu, \quad n = 0, 1, 2, \ldots$$

where $h = 6.626 \times 10^{-34}$ J·s is **Planck's constant**.

This led to Planck's radiation law:

$$\boxed{\rho_{\text{Planck}}(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{e^{h\nu/k_B T} - 1}}$$

Significance: First introduction of quantization, though Planck considered it a mathematical trick.



Key Features of Planck's Spectrum

- **Ultraviolet catastrophe**: Rayleigh-Jeans law diverges as ν^2 (orange curve)
- **Quantum resolution**: Planck's law remains finite due to $e^{h\nu/k_BT} - 1$ denominator
- **Wien's displacement law**: Peak frequency $\nu_{\text{max}} \propto T$
- **Stefan-Boltzmann law**: Total power $\int_0^\infty \rho(\nu, T)d\nu \propto T^4$
- **Temperature dependence**: Higher T (red dashed) $\rightarrow$ higher peak, shifts to higher ν

Planck's Radiation Low and the Big Bang: Quantum Mechanics Across Cosmic Scales

Remarkable connection: The cosmic microwave background (CMB)—the afterglow of the Big Bang—exhibits a perfect Planck blackbody spectrum at $T = 2.7255$ K. This provides:

- **Cosmological validation**: Quantum mechanics accurately describes the universe's oldest radiation
- **Experimental triumph**: COBE satellite measurements (1990) showed $< 0.03\%$ deviation from Planck's formula
- **Historical irony**: Planck's "mathematical convenience" describes radiation from 13.8 billion years ago!
- **Universal scope**: Same quantum principles apply from atomic to cosmological scales

The CMB spectrum represents the most precise blackbody measurement in nature, confirming quantum theory's cosmic validity.

Photoelectric Effect and Einstein's Light Quanta (1905)

Definition 25 (Photoelectric Effect)**.** *Emission of electrons from a metal surface when illuminated by light of sufficiently high frequency.*

Classical predictions vs. experimental observations:

Aspect	**Classical Prediction**	**Experimental Fact**
Dependence on intensity	Higher intensity $\Rightarrow$ higher KE	KE independent of intensity
Time delay	Energy accumulates over time	Emission instantaneous ($< 10^{-9}$ s)
Frequency threshold	No minimum frequency	Threshold frequency ν_0 exists

Einstein's Explanation (1905)

Light consists of discrete packets (photons) with energy:

$$E_{\text{photon}} = h\nu$$

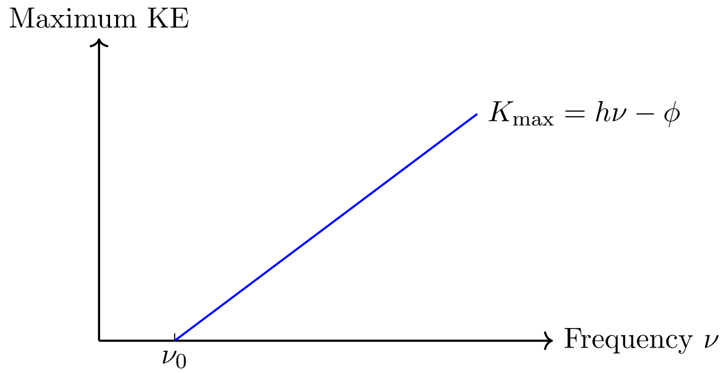
The maximum kinetic energy of emitted electrons:

$$K_{\max} = h\nu - \phi$$

where ϕ is the work function (minimum energy to remove electron).

Mathematical Formulation:

$$\frac{1}{2}m_e v_{\max}^2 = h\nu - \phi$$



Photoelectric Effect: Linear dependence of maximum kinetic energy on frequency

Significance: Light has particle-like properties; wave-particle duality begins.

5.1.2 Atomic Spectra and Bohr's Model (1913)

Atoms emit/absorb light at specific frequencies (spectral lines), not continuous spectra.

Example 3 (Hydrogen Spectrum)**.** *Balmer formula (1885) for visible lines:*

$$\frac{1}{\lambda} = R_H \left(\frac{1}{2^2} - \frac{1}{n^2} \right), \quad n = 3, 4, 5, \ldots$$

where $R_H = 1.097 \times 10^7 \ m^{-1}$ is the Rydberg constant.

Bohr's Postulates (1913)

1. Electrons orbit nucleus in **stationary states** with fixed energy (no radiation)
2. Radiation occurs only during **quantum jumps** between stationary states:

$$h\nu = E_i - E_f$$

3. Angular momentum is quantized:

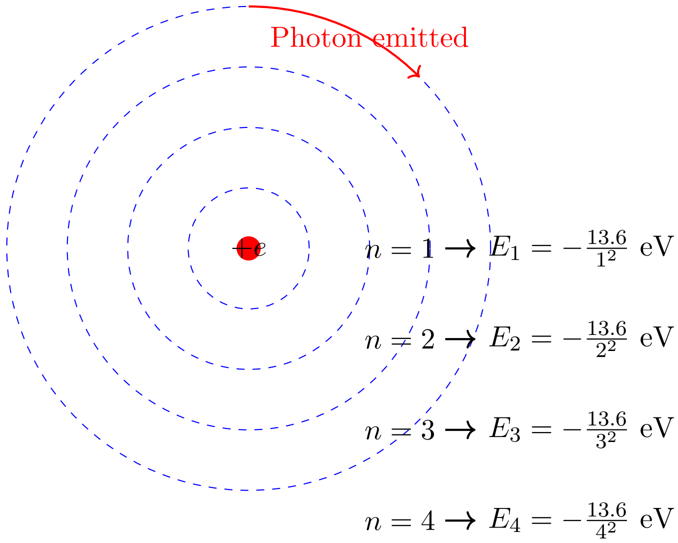
$$L = n\hbar, \quad n = 1, 2, 3, \ldots$$

where $\hbar = h/2\pi$.

From these postulates, Bohr derived for hydrogen atom:

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 h^2}\frac{1}{n^2} = -\frac{13.6 \text{ eV}}{n^2}$$

$$r_n = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}n^2 = a_0 n^2 \quad (a_0 = 0.529 \text{ Å})$$



Bohr Model: Quantized Orbits

Successes: Explained hydrogen spectrum, predicted Rydberg constant.

Limitations: Only worked for hydrogen-like atoms; no explanation for line intensities; ad hoc mixture of classical and quantum ideas.

5.2 Two Formulations of Quantum Mechanics

Two Equivalent Approaches to Quantum Mechanics

Historically, quantum mechanics developed through two parallel formulations that were later shown to be mathematically equivalent:

Aspect	**Wave Mechanics (Schrödinger)**	**Matrix Mechanics (Heisenberg)**
Key Figures	de Broglie, Schrödinger	Heisenberg, Born, Jordan
Fundamental Object	Wave function $\Psi(x,t)$	Infinite matrices X_{mn}, P_{mn}
Dynamics	Schrödinger equation: $i\hbar\frac{\partial\Psi}{\partial t} = \hat{H}\Psi$	Heisenberg equation: $\frac{d\hat{A}}{dt} = \frac{1}{i\hbar}[\hat{A},\hat{H}]$
Physical Interpretation	Probability amplitude	Transition amplitudes
Mathematical Framework	Differential equations	Linear algebra
Year Developed	1926	1925

Table 1: Comparison of the two historical formulations of quantum mechanics

Historical Unification

The equivalence of wave and matrix mechanics was established through:

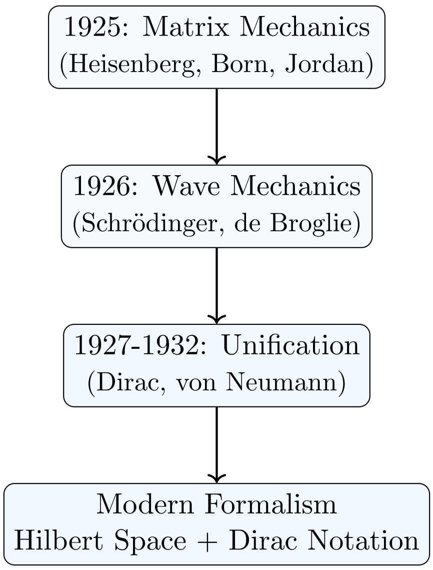
- **P.A.M. Dirac** (1927): Transformation theory showing equivalence
- **John von Neumann** (1932): Rigorous Hilbert space formulation
- **Unified framework**: Hilbert spaces with Dirac notation

Remark 5.1 (Historical vs. Modern Presentation)**.** *While historically matrix mechanics came first (Heisenberg, 1925), pedagogically we start with wave mechanics because:*

- ***Wave mechanics*** *is more intuitive for students familiar with differential equations*
- ***Visualizable concepts****: Wave functions, probability densities, interference*
- ***Direct connection*** *to classical wave physics*
- ***Easier calculations*** *for initial problems*

The modern Hilbert space formalism (combining both approaches) will be developed in outline once you have built physical intuition.

Historical Development of Quantum Mechanics



Our Pedagogical Approach

In this course, we will follow a **wave mechanics first** approach to build physical intuition before introducing abstract mathematical structures:

- **Part 1 (Lectures 2-6)**: Develop physical intuition through wave mechanics, the Schrödinger equation, and one-dimensional problems.
- **Part 2 (Lectures 7-10)**: Transition to the unified Dirac-Hilbert space formalism, introducing spin and abstract operators.
- **Part 3 (Lectures 11-14)**: Apply the complete quantum toolkit to advanced systems (harmonic oscillator, atoms, entanglement) and modern applications.

This progression allows us to ground the abstract postulates in concrete examples, mirroring the historical development of quantum theory.

What You Will Learn in This Course

Through our wave-mechanics-first approach, you will develop two complementary but equivalent skill sets:

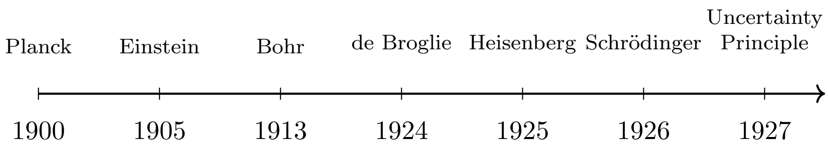
- **Wave Mechanics (Schrödinger Picture – Lectures 2–8)**:
  - Write and solve the Schrödinger equation for key potentials (infinite/finite wells, harmonic oscillator).
  - Physically interpret wavefunctions: $|\Psi(x,t)|^2$ as a probability density.
  - Calculate expectation values, uncertainties, and solve eigenvalue problems in position representation.
  - Understand dynamics through wave packets and the time evolution of probability densities.
- **Abstract Hilbert Space Formalism (Dirac Picture – Lectures 9–14)**:
  - Represent quantum states as abstract vectors $|\psi\rangle$ in Hilbert space.
  - Represent physical observables as operators (e.g., Pauli matrices for spin).
  - Master Dirac notation for calculations: $\langle\phi|\hat{A}|\psi\rangle$, $\sum_n |n\rangle\langle n| = \hat{I}$.
  - Solve problems using operator algebra (e.g., ladder operators for the harmonic oscillator).
  - Describe composite systems and entanglement using tensor products: $\mathcal{H}_A \otimes \mathcal{H}_B$.

Synthesis: By the end of the course, you will be able to translate physical problems into either formalism and appreciate their deep mathematical unity—the same physics expressed in different 'languages'.

Summary: Birth of Modern Quantum Mechanics (1925-1927)

Timeline of Quantum Revolution

Key Developments in Early Quantum Mechanics



Formulation	Creator	Key Idea
Matrix Mechanics	Heisenberg (1925)	Represent observables as matrices
Wave Mechanics	Schrödinger (1926)	Describe particles as waves $\Psi(x,t)$
Transformation Theory	Dirac (1927)	Unified both approaches
Probability Interpretation	Born (1926)	$|\Psi|^2$ as probability density
Uncertainty Principle	Heisenberg (1927)	$\Delta x \Delta p \geq \hbar/2$

6 The Double-Slit Experiment: Conceptual Foundation

Three Foundational Quantum Experiments

In this course, we will examine three pivotal experiments that reveal the core principles of quantum mechanics:

Experiment I: **Double-slit experiment**: Demonstrates wave-particle duality and quantum interference

Experiment II: **Stern-Gerlach experiment**: Confirms quantization of angular momentum (spin)

Experiment III: **EPR correlation experiment**: Reveals quantum nonlocality and entanglement

These experiments provide the empirical foundation upon which quantum theory is built.

6.1 Classical Wave Phenomena: Precursor to the Double-Slit Experiment

Before we analyze the quantum double-slit experiment, we must first understand classical wave phenomena. The most direct intuition can be developed through water waves, light waves, and their interference patterns. Richard Feynman's *Lectures on Physics* discusses this topic exceptionally well (Volume I, Chapters 51-52); here we review the essentials needed for quantum mechanics.

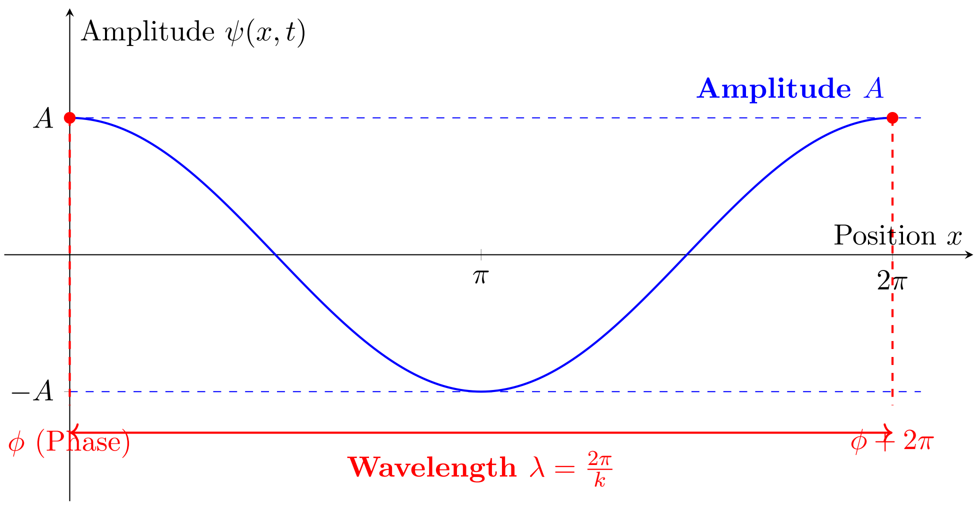
What is a Wave?

A wave is a disturbance that propagates through space and time, transferring energy without permanent displacement of the medium. In water waves, the medium is water molecules vibrating vertically; in light waves, oscillating electric and magnetic fields propagate even through vacuum.

Mathematical Description of Waves

Consider a simple one-dimensional wave propagating along the x-axis. The most general traveling wave can be written as:

$$\boxed{\psi(x,t) = A\cos(\omega t - kx + \phi)}$$



- **Amplitude** A: Maximum displacement from equilibrium (measured in meters for water waves, volts/meter for electric fields)
- **Angular frequency** ω: Rate of phase change with time ($\omega = 2\pi f$, units: rad/s)
- **Wave number** k: Rate of phase change with position ($k = 2\pi/\lambda$, units: rad/m)
- **Phase constant** ϕ: Initial phase at $x = 0, t = 0$

Phase Velocity and Dispersion Relation

The **phase velocity** v_p is the speed at which points of constant phase move. Setting the phase to constant:

$$\omega t - kx + \phi = \text{constant} \quad \Rightarrow \quad \omega\, dt - k\, dx = 0$$

$$\boxed{v_p = \frac{dx}{dt} = \frac{\omega}{k}}$$

The relationship $\omega = \omega(k)$ is called the **dispersion relation**:

- **Non-dispersive waves**: $\omega = vk$ (constant v), all frequencies travel at same speed (e.g., light in vacuum, sound in air)
- **Dispersive waves**: $\omega(k)$ non-linear, different frequencies travel at different speeds (e.g., water waves, light in materials)

Waves as Fields

Waves are Fields

A wave is an example of a **field**: a physical quantity defined at every point in space and time. Other important fields include:

- Electric field $\mathbf{E}(x, y, z, t)$ and magnetic field $\mathbf{B}(x, y, z, t)$
- Gravitational field $g(x, y, z, t)$
- Temperature field $T(x, y, z, t)$ (heat distribution)
- Quantum wave function $\Psi(x, y, z, t)$ (central to this course!)

This field concept will be crucial throughout quantum mechanics.

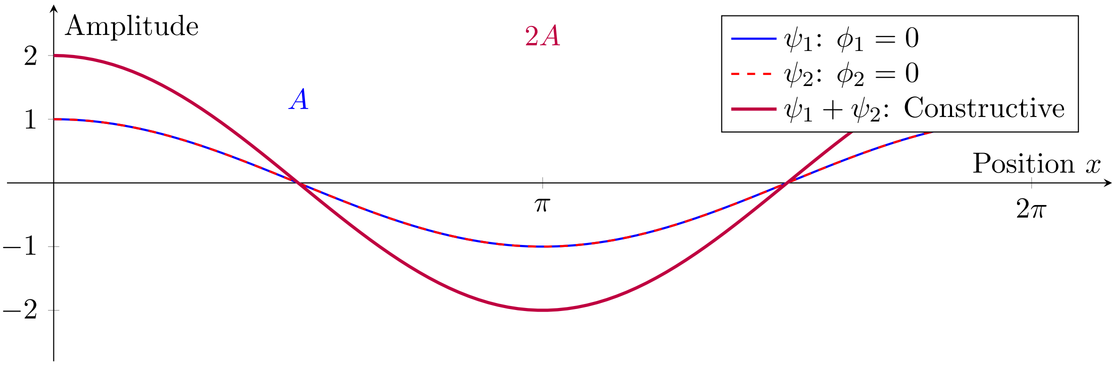
Interference: The Hallmark of Wave Behavior

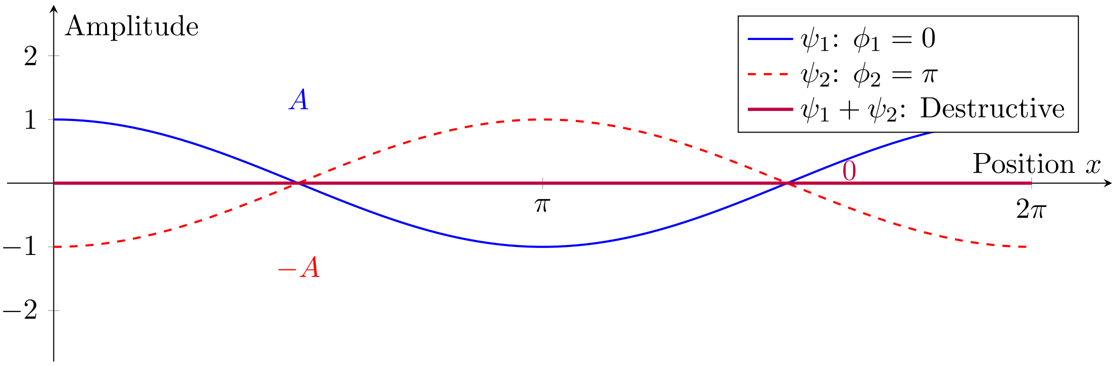
When two or more waves overlap, they **interfere**. Consider two waves with the same frequency but different phases:

$$\psi_{\text{total}}(x,t) = \psi_1(x,t) + \psi_2(x,t) = A\cos(\omega t - kx + \phi_1) + A\cos(\omega t - kx + \phi_2)$$

Using trigonometric identities, this simplifies to:

$$\psi_{\text{total}}(x,t) = 2A\cos\left(\frac{\phi_1 - \phi_2}{2}\right)\cos\left(\omega t - kx + \frac{\phi_1 + \phi_2}{2}\right)$$





- **Constructive interference**: $\phi_1 - \phi_2 = 2n\pi \Rightarrow$ amplitude doubles ($2A$)
- **Destructive interference**: $\phi_1 - \phi_2 = (2n+1)\pi \Rightarrow$ complete cancellation (0)
- **Partial interference**: Intermediate phase differences yield amplitudes between 0 and $2A$

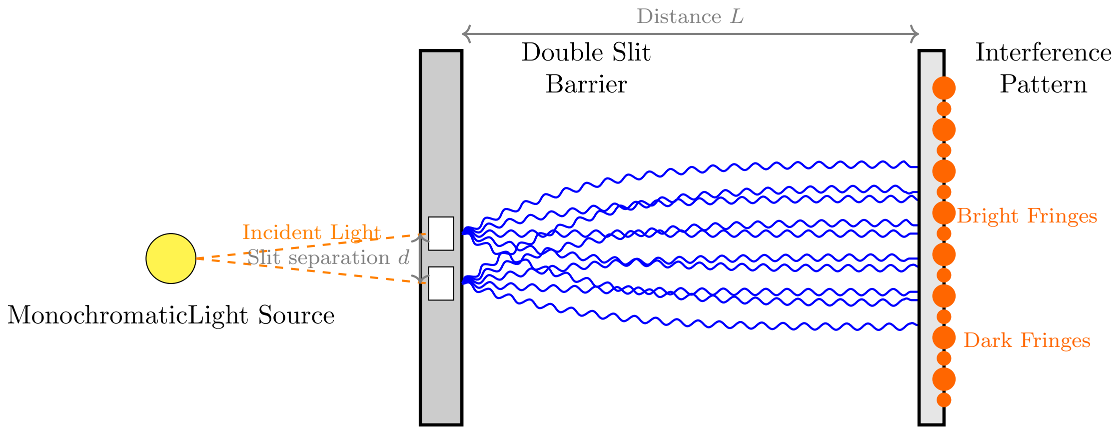
From Water Waves to Light: Young's Double-Slit Experiment (1801)

Thomas Young's Experiment

In 1801, Thomas Young performed the definitive experiment demonstrating light's wave nature:

- A monochromatic light source illuminates a barrier with two narrow slits
- Each slit acts as a new source of spherical waves (Huygens' principle)
- Waves from both slits interfere on a screen
- Result: Alternating bright (constructive) and dark (destructive) fringes

This proved light behaves as a wave, contradicting Newton's particle theory.



Young's Double-Slit Experiment (1801)

Demonstrated wave nature of light through interference fringes

Why This Matters for Quantum Mechanics

Bridge to Quantum Physics

The classical wave concepts developed here provide the essential foundation for understanding quantum phenomena:

- **Wave-particle duality**: Just as light waves show particle properties (photons), matter particles show wave properties (de Broglie waves)
- **Quantum interference**: The double-slit experiment with electrons produces identical patterns to light waves
- **Probability amplitudes**: Quantum wave functions $\Psi(x, t)$ obey superposition and interference like classical waves
- **Measurement**: The act of observation affects quantum systems analogously to how measuring a wave disturbs it

The mathematical formalism of waves—superposition, interference, and field description—directly carries over to quantum mechanics.

6.2 History of the Electron Double-Slit Experiment

Electrons were discovered in 1897 by J.J. Thomson through his cathode ray tube experiments. Thomson's work established that:

- Cathode rays consist of **negatively charged particles**
- These particles have a **specific charge-to-mass ratio** (e/m) much larger than atoms
- They are **constituents of all atoms**, making them the first identified subatomic particle
- They can be **emitted from metals** via thermionic emission when heated

Key Historical Milestones

The electron double-slit experiment evolved through theoretical proposals and gradual experimental realization:

Theoretical Foundation (1920s)

- **1924**: **Louis de Broglie** proposed matter waves with $\lambda = h/p$
- **1927**: **Clinton Davisson and Lester Germer** demonstrated electron diffraction from nickel crystals (Nobel Prize 1937)
- **1928**: **Max Born** first discussed the *thought experiment* of electrons through double slits in his papers on quantum mechanics

First Serious Proposal (1940s-1950s)

- **1949**: **Carlo Ferrari** (Italian physicist) published one of the first detailed theoretical analyses
- **1961**: **Claus Jönsson** (University of Tübingen, Germany) performed the **first actual electron double-slit experiment**

Jönsson's Experiment (1961)

Claus Jönsson achieved the first true electron double-slit interference using:

- 50 keV electrons ($\lambda \approx 0.05$ Å)
- Double slits in copper foil (width ~ 0.3 m, separation ~ 1 m)
- Electron microscope for detection
- Clear interference fringes observed

Published in *Zeitschrift für Physik* 161, 454 (1961).

Refinements and Single-Electron Experiments

- **1974**: **G. Möllenstedt and H. Düker** improved the experiment with biprism technique
- **1976**: **P.G. Merli, G.F. Missiroli, and G. Pozzi** published electron interference patterns
- **1989**: **Akira Tonomura** (Hitachi Labs) performed the **definitive single-electron double-slit experiment**

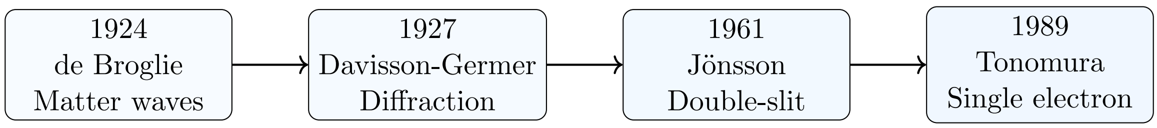
Tonomura's Single-Electron Experiment (1989)

Akira Tonomura's team demonstrated single-electron interference unambiguously:

- Electrons sent one at a time (spaced minutes apart!)
- Accumulated over time to build interference pattern
- Proved each electron interferes *with itself*
- Used electron biprism instead of physical slits
- Published in *American Journal of Physics* 57, 117 (1989)

This experiment conclusively demonstrated wave-particle duality at the single-particle level.

Evolution of Electron Interference Experiments



Key Scientific Implications

- **Jönsson (1961)**: First demonstration of electron interference through physical double slits
- **Tonomura (1989)**: Definitive proof of single-particle interference and wave-particle duality
- The experiments confirmed that:
  1. Electrons exhibit wave-like behavior (interference)
  2. Each electron interferes with *itself*, not with other electrons
  3. The wave function describes probability amplitudes
  4. Measurement collapses the wave function

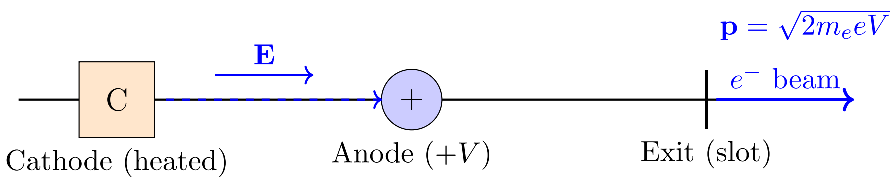
Remark 6.1 (Pedagogical Note)**.** *In textbooks, the "double-slit experiment" is often presented as:*

- *A **thought experiment** proposed by early quantum theorists (Born, Feynman)*
- *Actually performed in stages: Jönsson (1961) $\to$ improvements $\to$ Tonomura (1989)*
- *Richard Feynman called it "a phenomenon which is impossible [...] to explain in any classical way, and which has in it the heart of quantum mechanics"*

6.3 Double-Slit Experiment with Electrons: Phenomenological Description

The double-slit experiment represents perhaps the most fundamental demonstration of quantum mechanical principles. We will develop its understanding through several progressive stages.

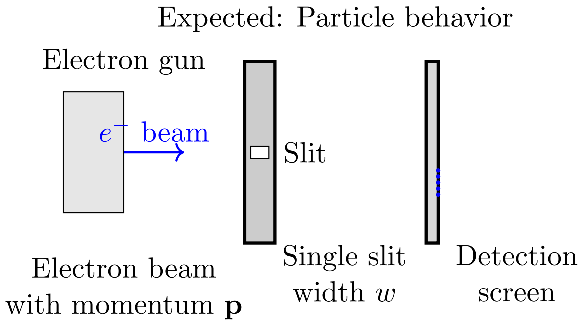
Stage 1: Electron Source Preparation



We begin with an **electron gun** constructed from a cathode ray tube (CRT):

- **Thermionic emission**: Electrons are detached from material by heating
- **Acceleration**: Electrons are accelerated by an electric field within the tube
- **Collimation**: Electrons exit through a narrow slot, producing a beam with controlled momentum $\mathbf{p}$
- **Result**: A monoenergetic electron beam with well-defined initial conditions

Stage 2: Single-Slit Preliminary Experiment

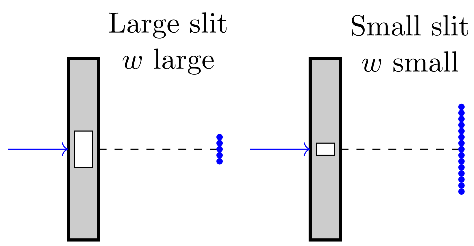


- **Setup**: The electron beam impinges on a barrier containing a single slit of width w
- **Detection**: A photographic screen records electron impacts as visible dots
- **Initial observation**: For sufficiently large slit width w, electrons form a distribution centered opposite the slit
- **Classical expectation**: Electrons as particles should pass through the slit and strike the screen directly opposite

Stage 3: The Quantum Surprise: Single-Slit Diffraction

Classical vs. Quantum Prediction

- **Classical particle prediction**: Decreasing slit width $w \to 0$ should produce *sharper* localization on screen
$$\Delta y_{\text{screen}} \propto w \quad \text{(smaller slit} \Rightarrow \text{tighter focus)}$$
- **Experimental observation**: Decreasing slit width produces *broader* distribution
$$\Delta y_{\text{screen}} \propto \frac{1}{w} \quad \text{(smaller slit} \Rightarrow \text{greater spread)}$$
- **Paradox**: The distribution *expands* as the slit narrows!



Wave Interpretation of Single-Slit Diffraction

The observed behavior finds natural explanation through wave optics:

- **Wave description**: Electrons exhibit wavelength $\lambda = h/p$ (de Broglie relation)
- **Diffraction principle**: A slit of width w acts as a secondary wave source
- **Angular spread**: $\Delta\theta \approx \lambda/w$ (smaller $w \Rightarrow$ larger $\Delta\theta$)
- **Screen spread**: $\Delta y_{\text{screen}} \approx L\Delta\theta \approx L\lambda/w$

- **Mathematical consistency**: The observed $\Delta y \propto 1/w$ matches wave prediction

Remark 6.2 (First Evidence of Wave Nature)**.** *The single-slit experiment provides* preliminary evidence *for the wave nature of electrons:*

1. *Electrons exhibit diffraction patterns characteristic of waves*
2. *The spreading obeys the mathematical relationship* $\Delta y \propto 1/w$
3. *This contradicts purely particle-like behavior*

However, this evidence remains ***inconclusive*** *because alternative particle-based explanations (e.g., interaction with slit edges) could potentially account for the observations.*

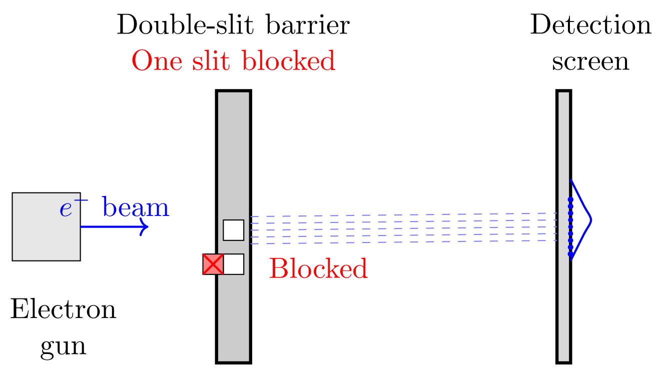
6.4 The Need for the Double-Slit Experiment

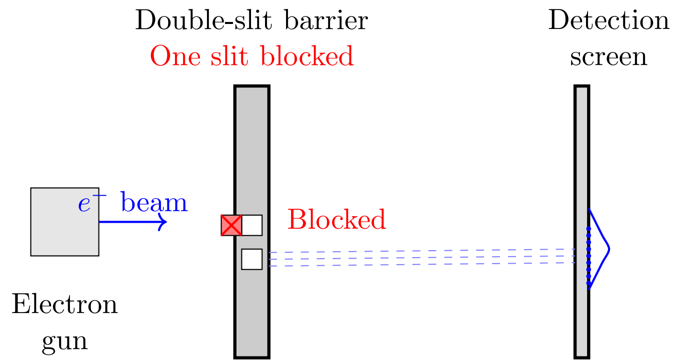
To obtain *definitive evidence* of wave behavior, we require an experiment that demonstrates **interference**—a phenomenon unique to waves:

Why Single-Slit is Not Conclusive

- Single-slit diffraction could potentially be explained by particle-slit interactions
- Particles might be scattered by the slit edges in a width-dependent manner
- Only interference patterns (requiring *two* coherent sources) provide unambiguous wave evidence
- The double-slit experiment eliminates alternative particle explanations

The double-slit configuration, which we will examine next, provides the decisive test by producing **interference fringes** that cannot be explained by any classical particle model. This interference pattern, with its characteristic maxima and minima, constitutes the definitive signature of wave behavior in quantum systems.





Single-Slit Control Experiment

When only one slit is open (the other blocked by a physical barrier), the experiment reduces to the single-slit case:

- **Electron source**: Monoenergetic beam from CRT gun
- **Barrier**: Double-slit apparatus with one slit physically blocked
- **Result**: Only electrons passing through the *unblocked slit* reach the screen
- **Distribution**: Pattern on screen shows **Gaussian-like spread** centered opposite the open slit

Summary of Single-Slit Results

Single-Slit Experiments: Particle-like Behavior

When we repeat the experiment with the **other slit blocked**, we obtain exactly the same type of results:

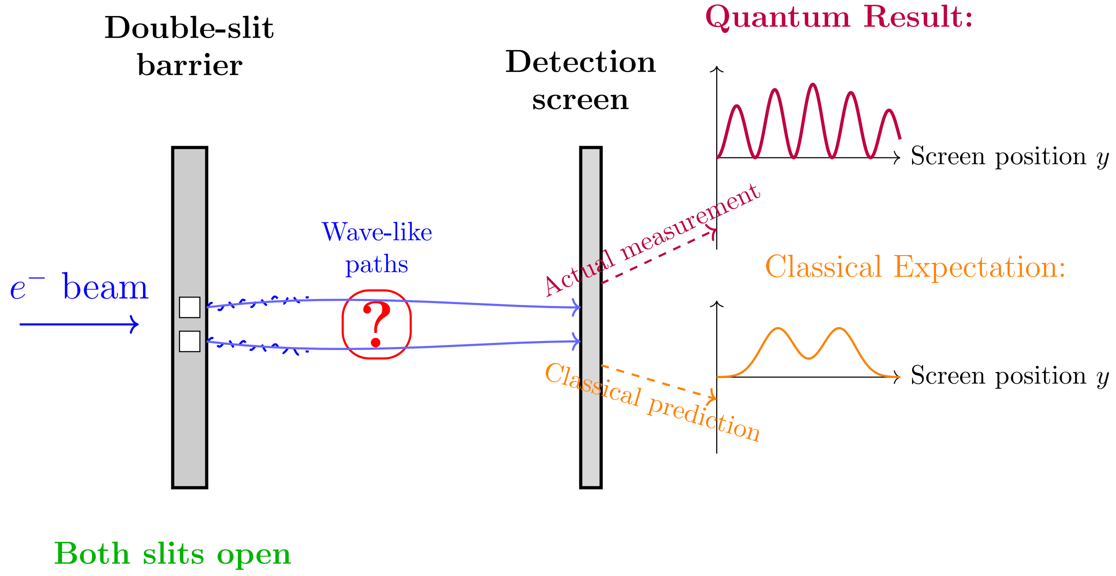
- With only **upper slit open**: Localized probability distribution centered opposite the upper slit
- With only **lower slit open**: Localized probability distribution centered opposite the lower slit

This indicates that electrons behave as **particles** in the sense that:

1. Each electron goes through a *specific slit* when only one is available
2. The arrival pattern follows a predictable probability distribution
3. Individual electrons make localized impacts on the screen

6.5 Transition to Double-Slit Experiment

The Critical Question: Both Slits Open



The Quantum Revelation

The side-by-side comparison reveals the fundamental departure from classical intuition:

- **Classical expectation**: Simple sum of independent probabilities

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- **Quantum reality**: Interference of probability amplitudes

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- **Critical difference**: The cross-term $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$
  - Can be **positive** (constructive interference → maxima)
  - Can be **negative** (destructive interference → minima)
  - Can be **exactly canceling** (zero probability regions!)

Paradox: Opening an *additional* path (second slit) creates regions where electrons *cannot* go—impossible in classical particle physics!

The Quantum Surprise Revealed

The interference pattern overlay shows what *actually* happens when both slits are open:

- **Not two Gaussians**: The pattern is *not* the sum of single-slit distributions
- **Interference fringes**: Alternating bright (maxima) and dark (minima) bands
- **Zero probability regions**: At minima, probability of detecting electrons is **exactly zero**
- **Quantum signature**: This pattern can only be explained by *wave interference*

$$\boxed{P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2 \neq |\psi_{\text{upper}}(y)|^2 + |\psi_{\text{lower}}(y)|^2}$$

The interference terms $\psi^*_{\text{upper}}\psi_{\text{lower}} + \psi_{\text{upper}}\psi^*_{\text{lower}}$ create the characteristic pattern.

The Fundamental Question Introduced

With both slits open, we face profound questions about the nature of quantum reality:

- **Particle perspective**: Does each electron go through *one specific slit*?
- **Wave perspective**: Does the electron pass through *both slits simultaneously* as a wave?
- **Measurement problem**: If we try to *observe* which slit the electron goes through, what happens?
- **Superposition**: Could the electron be in a *superposition* of both paths until measured?

The double-slit experiment with both slits open tests the very foundations of quantum mechanics.

Three Possible Interpretations

Classical Particle View
Each electron goes through
either upper *or* lower slit

$\xrightarrow{?}$

Wave View
Electron as wave passes through
both slits simultaneously

$\xrightarrow{?}$

Quantum Superposition
Electron in state describing
both possibilities at once

What the Experiment Will Test

- **If classical particle view is correct**: Pattern = Sum of two single-slit distributions

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

- **If wave/superposition view is correct**: Pattern = Interference of probability amplitudes

$$P_{\text{quantum}}(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$$

- **Critical difference**: The quantum prediction includes *interference terms*:

$$|\psi_1 + \psi_2|^2 = |\psi_1|^2 + |\psi_2|^2 + 2\text{Re}(\psi_1^* \psi_2)$$

The last term can be **negative**, creating regions of *zero probability*!

6.6 Fauliure of Classical Logical Analysis

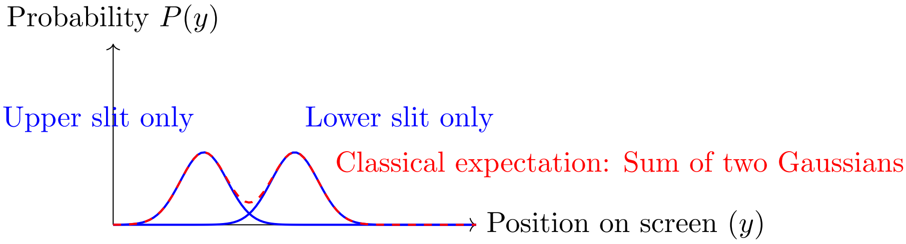
Classical Particle Expectation

Based on classical reasoning about particles, we expect the following:

- Electrons are particles $\Rightarrow$ each electron goes *either* through the upper slit *or* through the lower slit
- The logical operation is OR (union of two mutually exclusive events)
- Classical probability rule: For mutually exclusive events, total probability = sum of individual probabilities

$$P_{\text{classical}}(y) = P_{\text{upper}}(y) + P_{\text{lower}}(y)$$

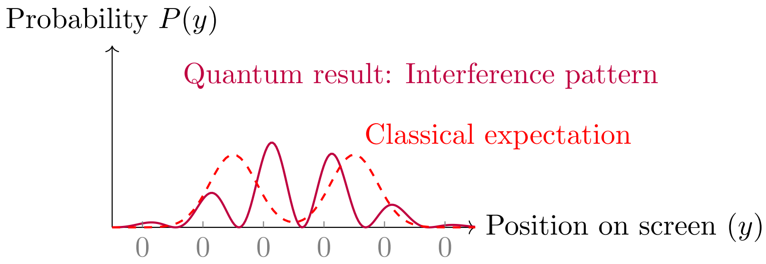
$$\boxed{P_{\text{classical}}(y) \propto \exp\left[-\frac{(y - y_{\text{upper}})^2}{2\sigma^2}\right] + \exp\left[-\frac{(y - y_{\text{lower}})^2}{2\sigma^2}\right]}$$



The Quantum Surprise: Interference Pattern

Experimental Result (Contrary to Classical Expectation)

When the experiment is actually performed with **both slits open**, we do *not* get the classically expected sum of two probability distributions. Instead, we observe a fundamentally different **interference pattern**:



Alternating maxima and **null regions** (zero probability!)

The Profound Implication

The interference pattern contains alternating **maxima** and **null regions**:

- **Null regions**: Probability of detecting electrons at these locations is *exactly zero*
- **Paradox**: When two slits are open, it becomes *impossible* for electrons to reach certain positions
- **Contradiction**: This *cannot* be explained by electrons going through either the upper OR lower slit
- If electrons were classical particles, opening an *additional* path should only *increase* probability, never create regions of *zero* probability!

6.7 Quantum Explanation: Wave-Particle Duality

Wave-Particle Duality and de Broglie Hypothesis (1924)

de Broglie's Insight

If light (waves) can behave as particles, perhaps matter (particles) can behave as waves:

$$\lambda_{\text{de Broglie}} = \frac{h}{p}$$

where p is the particle's momentum.

Evidence: Davisson-Germer experiment (1927) showed electron diffraction from crystals, confirming wave nature of electrons.

Significance: Complete symmetry between matter and radiation; foundation for wave mechanics.

Wave Mechanics Interpretation

The interference pattern can be explained if a **complex-valued wavelike field** is associated with the electron:

- **de Broglie-Schrödinger field**: $\psi(x, t)$ represents the quantum state
- **Wave nature**: The electron behaves as a wave passing through *both slits simultaneously*
- **Interference**: Waves from two slits interfere constructively (maxima) and destructively (nulls)
- **Probability**: $P(y) = |\psi_{\text{upper}}(y) + \psi_{\text{lower}}(y)|^2$ (not simple sum!)